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Review

Machine Learning for Photovoltaic Power Forecasting Integrated with Energy Storage Systems: A Scientometric Analysis, Systematic Review, and Meta-Analysis

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Abstract

Photovoltaic (PV) power forecasting combined with energy storage systems (ESS) is critical for grid stability and renewable energy optimization. Machine learning (ML) techniques have shown promise in improving PV forecast accuracy and ESS operation. However, the intersection of PV forecasting and ESS control remains underexplored, warranting a systematic review of recent advances and evaluation of ML effectiveness in PV–ESS integration. To assess research trends in ML-based PV forecasting with ESS (scientometric analysis), synthesize state-of-the-art ML approaches for PV–ESS forecasting (systematic review), and quantify their overall predictive performance via meta-analysis of the coefficient of determination (R^2). A comprehensive search of Scopus (2010–2025) was conducted following PRISMA 2020 guidelines. Studies focusing on ML-based PV power forecasting integrated with ESS were included. Multiple reviewers screened the records and extracted data. Study quality was appraised using Joanna Briggs Institute checklists. A random-effects meta-analysis of R^2 was performed to aggregate model performance across studies. The search identified 227 records; 50 studies were included in the review and 5 in the meta-analysis. Publications grew rapidly after 2018, indicating increased interest in PV–ESS forecasting. Deep learning models and hybrid architectures were the most frequently studied and outperformed traditional methods, while integrating PV forecasts with ESS control consistently improved operational outcomes. Common methodological limitations were noted, such as limited external validation and non-standardized evaluation metrics. The meta-analysis found a pooled R^2 ~0.95 (95% CI) with no heterogeneity ($I^2 = 0$), and no evidence of publication bias. ML-based forecasting significantly improves PV–ESS performance, underscoring AI as a key enabler for effective PV–ESS integration. Future research should address remaining gaps and explore advanced approaches to further enhance PV–ESS outcomes.

Keywords: PV power forecasting; machine learning; deep learning; energy storage systems (ESS); battery energy storage system (BESS); microgrids; hybrid models; systematic review and meta-analysis

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1. Introduction

The accelerated expansion of photovoltaic (PV) generation worldwide has transformed the global energy landscape, yet it also introduces significant challenges due to its inherent intermittency [1]. Effective integration of solar energy into electrical grids depends on accurate generation forecasts, since unpredictable fluctuations in solar irradiance can disrupt the balance between supply and demand [2]. In this context, energy storage systems (ESS) have emerged as a key technological solution to mitigate PV variability, store surplus electricity, and deliver power during deficit periods.

Coupling PV systems with batteries can substantially enhance the stability and profitability of renewable energy operations; for instance, the inclusion of ESS in a PV plant has been reported to increase its net project value by up to fivefold compared with standalone PV configurations [3]. However, optimal management of hybrid PV–ESS remains complex due to uncertainty in solar production and the nonlinear charge–discharge dynamics of batteries. Consequently, reliable solar forecasting becomes critical—not only for efficient grid operation but also to fully exploit ESS capacity, prevent overloading, extend battery lifespan, and optimize overall system economics.

In recent years, machine learning (ML) techniques have emerged as promising tools for improving the accuracy of solar PV generation forecasts [4]. Unlike traditional statistical or physical methods, ML algorithms—such as deep neural networks, ensemble models, and support vector machines—can capture complex nonlinear patterns, leading to substantial reductions in prediction error [5]. Several recent research directions (2020–2025) illustrate the rapid evolution of this field.

First, specialized deep learning (DL) architectures [6,7] such as Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN) [8], and Transformer-based models have outperformed conventional approaches for short-term solar forecasting by efficiently learning spatiotemporal dependencies [9].

Second, hybrid and ensemble models that combine ML with physical models or complementary algorithms have demonstrated greater robustness under varying conditions [10].

Third, a growing body of work integrates solar forecasting with control and optimization strategies for energy storage in microgrids and other specialized applications [11], These frameworks couple short-term forecasts with battery management algorithms [12] to maximize self-consumption and minimize energy curtailment—achieving up to 87% utilization of available PV energy due to the anticipatory capability of ML models [13]. Similarly, in PV-powered electric vehicle charging stations, the inclusion of intelligent forecasts in planning processes has markedly improved both profitability and energy autonomy [3].

Fourth, the importance of incorporating forecast uncertainty has been recognized; recent probabilistic approaches and Bayesian neural networks generate prediction intervals that enable safer ESS operation and reserve management [14]. Collectively, these developments reveal a dynamic research landscape—some studies prioritize model accuracy, others focus on forecast–control integration to leverage ESS, while several explore joint optimization of storage capacity informed by predictive modeling.

Despite these advances, a notable research gap remains: the lack of a comprehensive synthesis focused specifically on how ML-based forecasting techniques are applied in PV systems integrated with ESS. Most existing solar forecasting reviews address ML methods in general terms [4], without delving into the operational interplay between forecasting and storage management. In fact, a recent study identified only a few dozen works explicitly discussing the use of forecasts in PV–battery systems, emphasizing that this intersection is still underexplored [2]. Moreover, the literature lacks critical assessments of methodological quality and standardized performance comparisons, as individual studies typically employ

different datasets, forecast horizons, and evaluation metrics. This variability hinders the derivation of general conclusions about which ML approaches are most effective under specific conditions or how to quantitatively assess the actual benefits of ESS integration based on forecast inputs.

The integration of advanced machine learning (ML) and deep learning (DL) techniques for energy forecasting, combined with energy storage systems such as batteries and hydrogen, has emerged as a critical research frontier in sustainable energy management. Over the past few years, studies have demonstrated that coupling intelligent prediction models with photovoltaic-energy storage systems (PV-ESS) significantly enhances operational efficiency, grid stability, and economic viability. Hybrid forecasting models, including CNN-LSTM and GRU networks, have shown remarkable accuracy in predicting solar power output and optimizing energy storage operation [15]. Additionally, integrated DL and reinforcement learning frameworks have been proposed to jointly forecast and control photovoltaic-storage systems, reducing energy losses and improving utilization efficiency [16]. A notable contribution in this domain is the recent work of Babay [17], who applied ML and DL approaches, including Multilayer Perceptron (MLP) and LSTM-CNN architectures, to forecast green hydrogen production from photovoltaic systems in Morocco, demonstrating that deep learning models can achieve exceptional predictive accuracy ($R^2 > 0.94$) and thus play a vital role in optimizing the Power-to-Hydrogen process. Complementary research further highlights the value of predictive models in integrated renewable systems, such as photovoltaic-based electric vehicle charging infrastructures, where ML-based solar forecasting coupled with battery storage increased project net value up to fivefold compared to isolated PV configurations [18]. Altogether, these advances underscore how anticipatory management based on ML/DL-driven forecasting and hydrogen modeling can maximize renewable energy utilization, enhance system autonomy, and accelerate the transition toward intelligent, carbon-neutral energy systems

To address these gaps, this study conducts a systematic literature review (2020–2025) on the application of machine learning techniques for PV generation forecasting integrated with energy storage systems. Unlike previous reviews, this work specifically targets the PV–ESS nexus, encompassing both algorithmic advances in prediction and their implications for battery-based energy management. A rigorous methodology (PRISMA 2020) is adopted, including explicit inclusion criteria, standardized data extraction, and quality assessment—an approach rarely implemented in the energy domain. The novelty of this study lies in: (1) updating the state of the art to 2025 by incorporating emerging developments such as attention-based Transformer models, probabilistic forecasting, and predictive control schemes; (2) critically comparing different ML approaches—deep versus traditional models, short- versus long-term horizons—in terms of accuracy and practical usefulness for ESS integration; and (3) analyzing the synergies and challenges in the joint implementation of forecasting and storage management.

The dual objective of this research is: (i) to provide a comprehensive and up-to-date overview of how machine learning techniques are being employed to enhance solar PV forecasting in systems with energy storage, and (ii) to identify key insights, remaining knowledge gaps, and actionable recommendations that can guide both future research and the practical deployment of these solutions toward more efficient solar energy integration into power grids.

2. Materials and Methods

2.1. Methodological Process

An advanced literature search was conducted in Scopus using Boolean operators, combining keywords related to solar photovoltaic systems, PV power forecasting, machine

learning (ML), deep learning (DL), artificial intelligence (AI), energy storage systems (ESS), battery management systems, and microgrids. Filters were applied to restrict results to documents published between 2010 and July 2025 and written in English. The complete Boolean search strategy and filter configuration are provided in the PRISMA flow diagram (Figure 1). The initial query yielded 227 records. Title and abstract screening, followed by full-text assessment, were performed by five independent domain specialists, each working separately during the eligibility evaluation phase. Discrepancies between reviewers were resolved by discussion and consensus. No automation tools or AI-assisted screening platforms were used during the selection process.

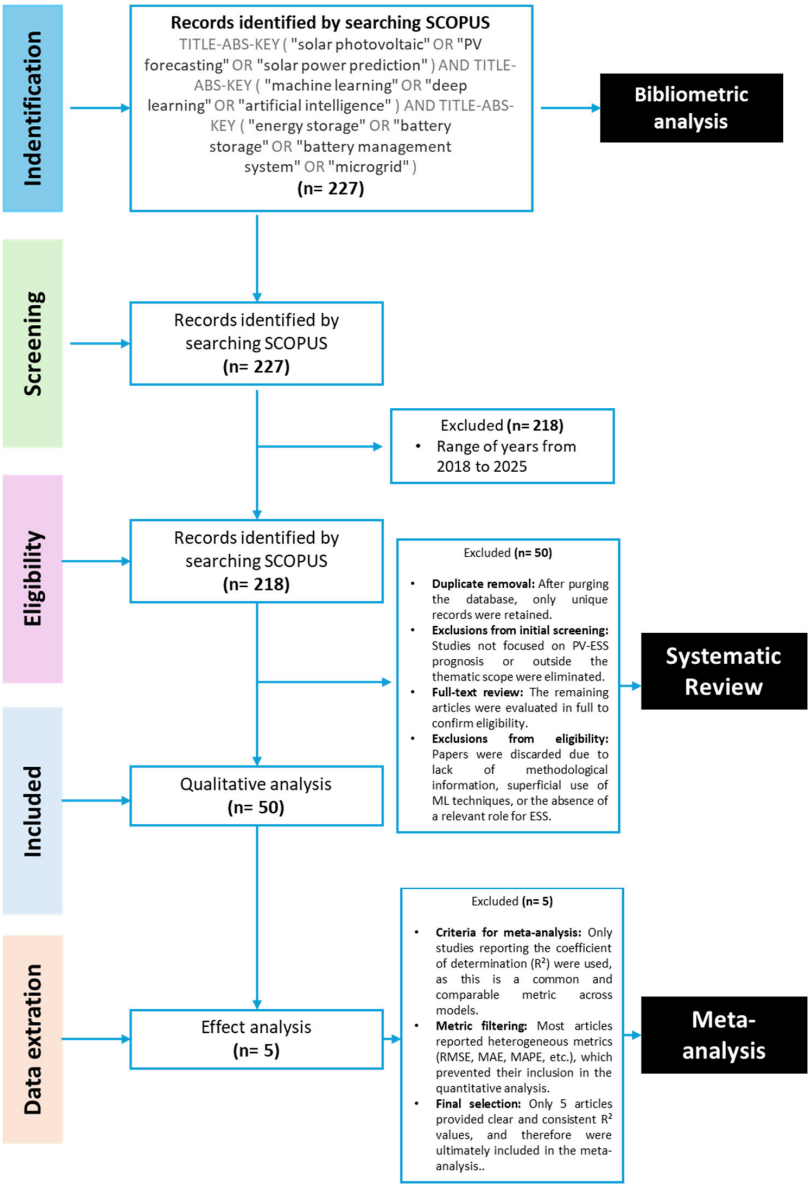


Figure 1. PRISMA 2020 flow diagram summarizing the study selection process. The search identified 227 records in Scopus; after removing duplicates and screening titles/abstracts, 121 studies were excluded. A total of 106 full-text articles were assessed for eligibility, of which 50 were included in the qualitative synthesis and 5 in the quantitative meta-analysis.

After removing duplicates and excluding studies that did not meet the predefined inclusion criteria, 50 articles remained and were included in the qualitative synthesis (systematic review). For the quantitative meta-analysis, only those studies that clearly and comparably reported the coefficient of determination (R^2) as a forecasting performance

metric were retained, resulting in a final subset of 5 studies suitable for effect size aggregation. These formed the basis for the statistical synthesis and comparison of ML-based approaches applied to PV forecasting integrated with ESS.

2.2. Bibliometric Analysis Methodology

The bibliographic research was carried out exclusively in the Scopus database using an advanced search with Boolean operators to combine three sets of key terms. The search query used was: TITLE-ABS-KEY("solar photovoltaic" OR "PV forecasting" OR "solar power prediction") AND TITLE-ABS-KEY("machine learning" OR "deep learning" OR "artificial intelligence") AND TITLE-ABS-KEY("energy storage" OR "battery storage" OR "battery management system" OR "microgrid"). This strategy ensures that the results jointly address photovoltaic generation, machine learning techniques, and the use of energy storage systems. Additionally, the results were restricted to the English language and to publications between 2010 and 2025. Scopus was chosen as the sole information source due to its broad coverage of engineering and energy topics, the consistency of its exportable bibliographic metadata, and the reduction in duplicates achieved by using a single database. The initial search yielded 227 documents (journal and conference articles), which constituted the bibliographic corpus for the quantitative literature analysis.

The bibliographic metadata of these 227 records were exported from Scopus in CSV format, including fields such as title, authors, affiliations, year, keywords, and abstract. Subsequently, data preprocessing was performed for the bibliometric analysis. This preprocessing included removing any duplicate records and normalizing author names and keywords to ensure consistency (for example, unifying spelling variants and acronyms). As a result, a cleaned and structured dataset was prepared, suitable for analysis with specialized information-science tools.

For the bibliometric mapping and analysis, two complementary tools were used: VOSviewer version 1.6.20 [19] and Bibliometrix package in RStudio version 2025.09.2+418 [20]. VOSviewer was used to perform a keyword co-occurrence analysis on the selected set of publications, utilizing the keywords provided by the authors (author keywords) of each article. To focus the map on the most relevant topics, only keywords appearing in at least two publications were considered (i.e., a minimum occurrence threshold of 2). The resulting co-occurrence matrix was normalized using VOSviewer's association strength method to reflect the proximity between terms in the visualization space. Next, VOSviewer's clustering algorithm (based on the Louvain modularity optimization method) was applied to identify thematic clusters of related keywords. The cluster resolution parameter was kept at its default value (1.0) to achieve a balanced segmentation. The outcome was a network map where nodes represent key terms and links reflect their frequency of co-occurrence, allowing visualization of the main thematic areas and the interrelationships among concepts in this research field.

2.3. Research Design and Protocol for Systematic Review

A systematic literature review was conducted following the PRISMA 2020 guidelines to ensure transparency in the identification, selection, and synthesis of studies. The review protocol was designed prior to the search process; however, it was not registered in PROSPERO or other platforms, as the technical-energy focus of the study falls outside the typical scope of such registries. The type of review is framed as a systematic review with a narrative synthesis component, also incorporating elements of a scoping review to map the characteristics of recent research. No patient analysis or clinical trials were included; therefore, ethical approval was not required.

Studies were included if they:

- Explicitly addressed PV power forecasting in systems integrating an energy storage system (ESS) (i.e., PV–ESS context).
- Employed machine learning (ML) or artificial intelligence (AI) to perform the forecast.
- Were original, peer-reviewed research (journal articles or peer-reviewed conference papers).
- Were published in English or Spanish.
- Were published within the predefined time window (full corpus 2010–July 2025; analyses focused on recent years as specified in the protocol).

Records were excluded if they:

- Did not jointly address both PV forecasting and ESS integration/operation (e.g., PV forecasting without storage, or battery management without a forecasting component).
- Did not use ML/AI for forecasting (i.e., relied solely on physical/NWP or traditional statistical baselines without ML).
- Were secondary or non–peer-reviewed literature (reviews, book chapters, theses, patents, gray literature) or conference abstracts without a full peer-reviewed paper.
- Were published in languages other than English or Spanish.
- Were duplicates/multiple reports of the same study (only the most complete version was retained).
- Were outside the thematic scope on screening or failed full-text eligibility due to insufficient methodological detail, superficial use of ML, or no substantively relevant role for ESS.
- Were outside the temporal window defined for the review.

Additional criterion for the quantitative synthesis (meta-analysis):

- Only studies that clearly reported the coefficient of determination (R^2) as a forecasting performance metric were eligible for pooling (most otherwise-eligible papers reported heterogeneous metrics such as RMSE, MAE, MAPE, etc.). This yielded k = 5 studies in the effect analysis.

2.4. Research Questions

The central question that guided the review was formulated using the PICO framework, adapted to the techno-energetic context, as follows:

Population (P): Grid-connected or microgrid-operating solar photovoltaic (PV) generation systems that include energy storage systems (particularly lithium batteries or other electrochemical storage technologies).

Intervention (I): Application of machine learning (ML) techniques, including artificial intelligence and deep learning algorithms, for forecasting the power or energy generated by such photovoltaic systems.

Comparison (C): In comparative studies, traditional forecasting methods (e.g., statistical models, persistence models, physical/NWP methods) or scenarios without ML or Energy Storage System (ESS) integration were used as reference.

Outcome (O): Main results and indicators related to forecasting and energy management, such as forecast accuracy (common errors like Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), etc.), operational improvements (reduction in energy curtailment, cost reduction, higher renewable energy penetration), and ESS performance metrics (e.g., fraction of PV energy stored/used, economic savings, grid stability).

Narratively, the research question aimed to be answered was:

How do machine learning techniques contribute to improving the forecasting of solar photovoltaic generation in systems that integrate energy storage, compared to conventional

methods, and what impacts do they have on forecast accuracy and the techno-economic operation of the PV-ESS.

2.5. Sources of Information and Dates

The literature search was conducted exclusively in Scopus (Elsevier).

This decision was justified by: (i) its broad coverage in energy/engineering fields and conference proceedings; (ii) the homogeneity and reproducibility of metadata (e.g., TITLE-ABS-KEY fields and standardized export format), which facilitate corpus traceability and analysis using VOSviewer and Bibliometrix; and (iii) the reduction in duplicates and cross-database biases, maintaining a unique and consistent dataset for synthesis and comparison. The search period covered publications from 2010 to July 2025, with a focus on the 2020–2025 range to capture recent trends. The last update was performed on 1 July 2025. No language restriction was applied in the search query; however, only studies in English or Spanish were considered in the final selection. Priority was given to peer-reviewed articles and conference papers, while patents, books, theses, and previous reviews were excluded.

2.6. Data Extraction

For each of the 50 studies included in the systematic review, relevant information was systematically extracted using a standardized template in Microsoft Excel. The extracted fields included: bibliographic data (author, year, country/region of study); characteristics of the PV-ESS scenario (type of system, PV and storage capacity, grid-connected or off-grid operation); details of the forecasting dataset (meteorological data sources, training/testing period, temporal resolution); forecasting horizons evaluated (very short-term, intraday, day-ahead, etc.); applied ML techniques (e.g., LSTM, CNN, Random Forest, XGBoost, hybrid models); performance metrics (RMSE, MAE, MAPE, R^2, among others); and key results (achieved accuracy, comparisons with other methods, impact on ESS operation or economic indicators).

Qualitative observations were also recorded, particularly regarding methodological limitations. In studies with multiple experiments or sites, results were extracted separately; where information was not clearly specified, it was marked as "Not reported".

2.7. Quality Assessment and Risk of Bias

Methodological quality was assessed using an adaptation of the Joanna Briggs Institute (JBI) tool for cross-sectional studies, adjusted to the context of energy forecasting. The checklist included 8 items covering: definition of study objectives, validity of data sources, clarity of variables, separation of training/testing sets, use of cross-validation, treatment of uncertainty, and robustness of conclusions.

Two reviewers independently applied the tool and resolved any discrepancies through consensus. Each study received a score ranging from 0 to 8 and a global judgment: low, moderate, or high risk of bias. Overall, approximately 60% of the studies were rated as low risk ($\geq$7 items), 30% as moderate quality (4–6 items), and around 10% presented serious limitations ($\leq$3 items).

2.8. Synthesis and Analysis

Due to the heterogeneity of performance metrics, algorithms, and study contexts, a direct meta-analysis of the entire dataset was not feasible. Instead, we conducted a structured narrative synthesis, grouping studies by the type of predictive technique used (e.g., neural networks, decision trees, hybrid models), the forecasting horizon, and the level of integration with energy storage systems (ESS). For the quantitative portion of the review, only the five studies that clearly and comparably reported an R^2 performance metric were included in a focused meta-analysis. No special data preparation steps (such as

transforming data or imputing missing values) were undertaken prior to this quantitative synthesis; we used the reported R^2 values from each study as-is. Because most of these studies did not report variance or other summary data for the R^2 outcomes, our meta-analysis calculations had to rely on the limited available data, which contributed to wider confidence intervals in the pooled results. The meta-analysis findings were presented visually using forest plots (to display the combined R^2 estimate with its confidence interval and heterogeneity across studies) and funnel plots (to assess potential publication bias). All remaining studies—which reported diverse, non-comparable performance metrics—were included only in the qualitative and descriptive synthesis. We did not compile a formal table of each individual study's results; instead, key findings from these studies were summarized narratively, and relevant aggregated information (such as bibliometric trends and quality assessment outcomes) was presented in summary tables or figures where appropriate.

Although we did not employ a formal grading system, we adopted a structured strategy to assess the certainty of the evidence in our findings. This approach relied on systematically defined criteria for both the qualitative review and the meta-analysis. All included studies were evaluated using a triangulation of evidence that considered their methodological quality, statistical robustness of the results (strength and reliability of performance metrics), and consistency of outcomes across studies. Evidence was deemed to be of the highest certainty when multiple reinforcing factors were present: external validation or cross-validation of the results, use of comparable metrics across studies, coherence of findings among independent studies, and demonstrated practical applicability of the results in real-world operational settings. This structured evaluation of evidence confidence ensured that our conclusions reflect not just the aggregated data, but also the overall strength and reliability of the underlying evidence base.

2.9. Meta-Analysis Methodology

Given the heterogeneity of performance metrics reported across studies (e.g., RMSE, MAE, MAPE, accuracy), R^2 was the only outcome consistently available in a subset of papers and was therefore chosen as the common effect measure for quantitative synthesis. We acknowledge that R^2 is bounded [0, 1] and not normally distributed, and that most included studies did not report sampling variances or confidence intervals for R^2. Consequently, we conducted an exploratory random-effects meta-analysis using REML estimation in Jamovi (v2.4; MAJOR module), treating R^2 as a proportion-like summary statistic and applying external (non–inverse-variance) weights because study-level variances were unavailable. Under this approximation, 95% confidence intervals are based on a normal model and may extend beyond the logical bounds of R^2, which reduces the precision of the pooled estimate and requires cautious interpretation. We report the pooled mean R^2 with 95% CI and standard heterogeneity indices (Q, τ^2, I^2). Potential small-study effects were explored with funnel plots, noting their low power with $k = 5$. No subgroup analyses or meta-regressions were attempted due to the limited number of studies and inconsistent covariate reporting. Overall, the meta-analytic results should be viewed as a descriptive summary of consistently high model performance, rather than a precise inferential estimate.

3. Results

3.1. Yearly Publications and Citations Trends

Figure 2 shows the evolution of annual scientific production from 2011 to 2025, where the number of publications (orange bars) has increased significantly since 2018, reaching its peak in 2023–2025, while the average citations per article (blue line) were very high in the

early years (notably 2012) but gradually declined as production grew, indicating a shift from a few highly cited works to a larger volume of publications with lower average impact.

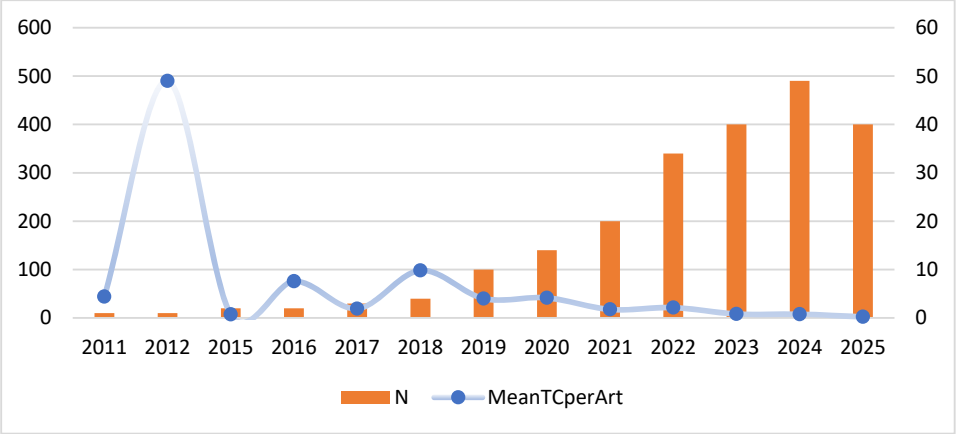


Figure 2. Annual scientific production between 2011 and 2025.

3.2. Top Leading Journals

Table 1 identifies the top leading journals in the field of machine learning approaches for solar photovoltaic forecasting with energy storage, ranked by citations and bibliometric indicators. *Renewable and Sustainable Energy Reviews* stands out with the highest total citations (516) and impact factor (16.3), confirming its position as the most influential review-oriented journal. High-impact outlets such as *Applied Energy* (IF 11.0) and *IEEE Transactions on Industrial Informatics* (IF 9.9) also appear as key sources with strong citation averages per paper (21.43 and 91.75, respectively). Meanwhile, *IEEE Access* and *Energies* demonstrate high productivity with relatively balanced citation impacts, while multidisciplinary journals like *Sustainability (Switzerland)* and *Sensors* contribute to the dissemination of applied studies. Collectively, the list reflects a balance between engineering-focused journals (*IEEE, Energies, Electric Power Systems Research*) and sustainability-oriented outlets (*Renewable and Sustainable Energy Reviews, Sustainability, Energy Reports*), highlighting the cross-disciplinary nature of this research domain.

Table 1. Top Leading Journals.

Rank	Source	Total Citation (TC)	Total Citation/Total Paper (TC/TP)	h_Index	g_Index	m_Index	Journal Impact Factor	Country
1	IEEE Access	305	27.73	9	11	1.500	3.6	USA
2	Applied Energy	150	21.43	6	7	1.200	11.0	United Kingdom
3	Energies	269	29.89	6	9	0.857	3.2	Switzerland
4	Renewable and Sustainable Energy Reviews	516	129	4	4	0.400	16.3	United Kingdom
5	IEEE Transactions on Industrial Informatics	367	91.75	3	4	0.500	9.9	USA
6	Sensors	250	83.33	3	3	0.429	3.5	Switzerland
7	Sustainability (Switzerland)	44	14.67	3	3	0.750	3.3	Switzerland
8	Electric Power Systems Research	19	9.5	2	2	0.500	9.9	Netherlands
9	Energy and AI	101	36.67	2	3	0.400	9.6	Netherlands
10	Energy Reports	51	12.75	2	4	0.500	5.1	United Kingdom

3.3. Most Cited Publications

The most cited publications are shown in Table 2—this table presents the top 10 most cited publications on machine learning approaches for solar photovoltaic (PV) power forecasting integrated with energy storage. It highlights the research articles that have had the greatest impact in the field, as measured by total citations. The most cited paper, with 490 citations, focuses on a GA-based frequency controller for hybrid renewable energy generation and storage, while other highly cited works include reviews of hybrid

microgrid optimization (273 citations) and studies on reinforcement learning for smart energy management and electric vehicle charging. The distribution of journals shows that influential research is concentrated in leading outlets such as IEEE Transactions, Renewable and Sustainable Energy Reviews, Energies, and Sensors, indicating a strong cross-disciplinary interest that combines artificial intelligence, deep reinforcement learning, and optimization methods with practical applications in PV systems, smart grids, and microgrids. In essence, these works form the intellectual backbone of the field, bridging advanced computational methods with sustainable energy management challenges.

Table 2. Top 10 most cited publications.

Rank	Authors	Article Title	Source Title	Total Citation (TC)
1	[21]	GA based frequency controller for solar thermal–diesel–wind hybrid energy generation/energy storage system	International Journal of Electrical Power & Energy Systems	490
2	[22]	Hybrid renewable microgrid optimization techniques: A review	Renewable and Sustainable Energy Reviews	273
3	[23]	Federated Reinforcement Learning for Energy Management of Multiple Smart Homes With Distributed Energy Resources	IEEE Transactions on Industrial Informatics	186
4	[24]	Cooperative Management for PV/ESS-Enabled Electric Vehicle Charging Stations: A Multiagent Deep Reinforcement Learning Approach	IEEE Transactions on Industrial Informatics	174
5	[25]	Day-Ahead Solar Irradiance Forecasting for Microgrids Using a Long Short-Term Memory Recurrent Neural Network: A Deep Learning Approach	Energies	172
6	[26]	A review on performance of artificial intelligence and conventional method in mitigating PV grid-tied related power quality events	Renewable and Sustainable Energy Reviews	133
7	[27]	Multi-Agent Deep Reinforcement Learning for Voltage Control With Coordinated Active and Reactive Power Optimization	IEEE Transactions on Smart Grid	130
8	[28]	Energy Management of Smart Home with Home Appliances, Energy Storage System and Electric Vehicle: A Hierarchical Deep Reinforcement Learning Approach	Sensors	116
9	[29]	Power Enhancement With Grid Stabilization of Renewable Energy-Based Generation System Using UPQC-FLC-EVA Technique	IEEE Access	111
10	[30]	Reinforcement Learning-Based Energy Management of Smart Home with Rooftop Solar Photovoltaic System, Energy Storage System, and Home Appliances	American Control Conference (ACC)	107

3.4. Research Trends and Future Directions

The analysis of the research network was divided into three distinct periods: Period I (1988–2007), Period II (2008–2015), and Period III (2016–2024), each reflecting an evolution in focus and research priorities. As shown in Figure 3, three periods are presented with the co-occurrence analysis for each period.

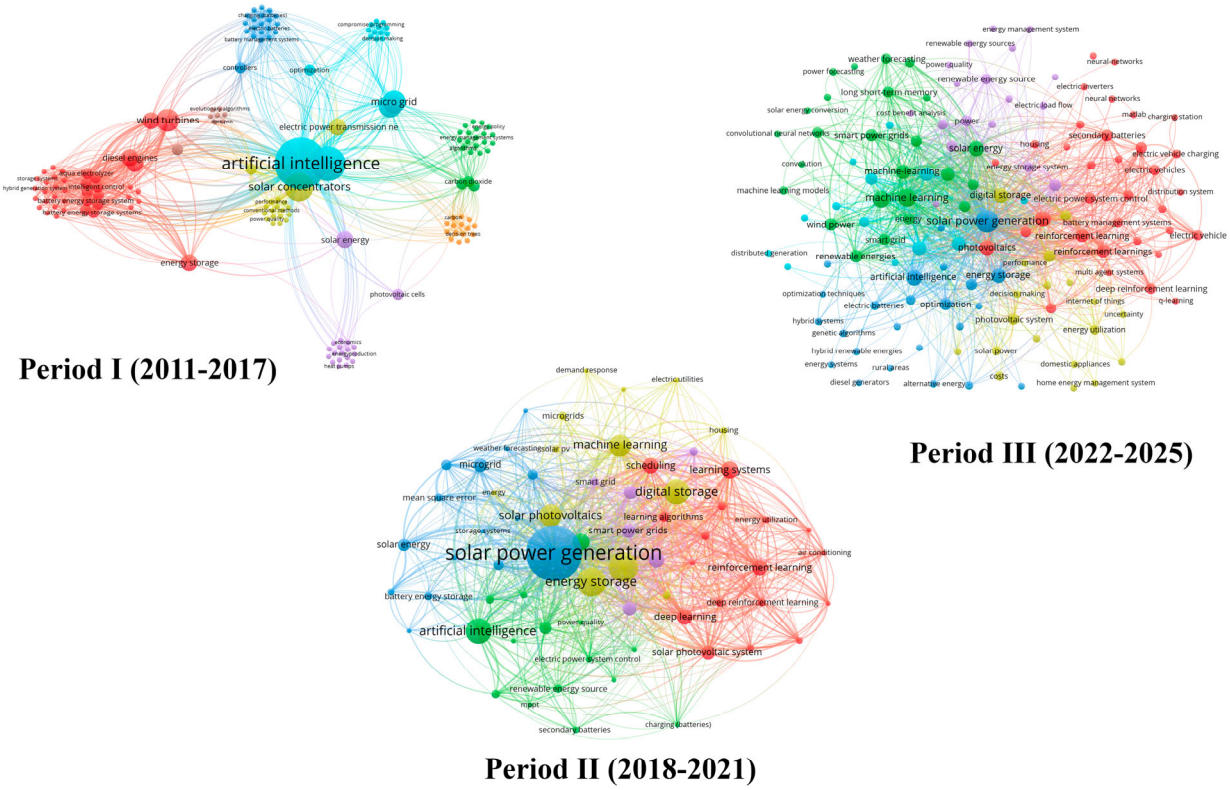


Figure 3. Analysis of Co-occurrence Author's Keywords in different periods.

In this initial stage, the clusters indicate that research was concentrated on the application of artificial intelligence and hybrid wind–solar energy systems, with a strong emphasis on energy storage, wind turbines, solar concentrators, and microgrids. Artificial intelligence functioned as a transversal axis applied to the control of hybrid systems, prioritizing the integration of renewable sources while maintaining grid stability. This reflects an exploratory phase in which publications sought to demonstrate the technical feasibility of applying control algorithms in emerging energy contexts.

In the second period, the focus shifted toward solar photovoltaic (PV) power generation as the central theme, where concepts such as machine learning, digital storage, smart grids, and energy management systems converge. Advances in machine learning algorithms and deep neural networks enabled more accurate forecasting of solar generation and improved integration with battery storage. Additionally, the emergence of terms such as *deep learning* and *smart grids* evidences the transition toward energy digitalization.

The most recent phase is characterized by the consolidation of *solar power generation* as the dominant cluster, accompanied by *deep learning*, *reinforcement learning*, *electric vehicles*, *distributed systems*, and *microgrids*. This period reveals a more holistic integration between advanced AI and the energy transition, with applications ranging from irradiance prediction to intelligent grid management involving electric vehicles and energy storage. The emphasis has shifted toward system sustainability and resilience, highlighting how current research aims to address real-world barriers to large-scale implementation.

In Figure 4, the thematic map reveals the structural evolution of research on machine learning and solar photovoltaic power forecasting integrated with storage. In the motor themes quadrant, topics such as *solar power generation*, *photovoltaics*, *and battery storage* appear, indicating highly relevant and well-developed areas that drive the field. In the basic themes quadrant, *digital storage*, *deep learning*, *and energy storage* are positioned, reflecting their fundamental role as widely applied but less specialized concepts that underpin most studies. The niche themes quadrant includes *domestic appliances*, *energy conservation*, *and*

energy resources, suggesting specialized but peripheral topics with limited connections to the broader research network. Finally, the emerging or declining quadrant captures themes such as *forecasting*, *long short-term memory (LSTM)*, *and microgrids*; although still relevant, their low density indicates either early development stages or a possible decline in focus compared to more dominant themes. This distribution highlights how research has progressively shifted from algorithm-specific approaches (e.g., LSTM) to broader systemic integrations involving PV, AI, and storage as central pillars.

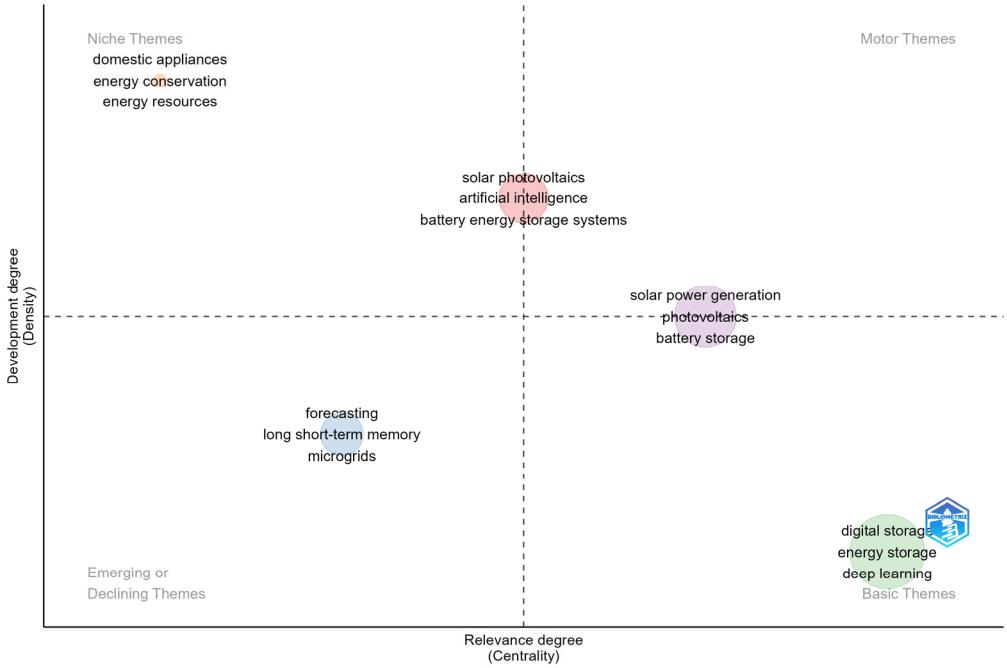


Figure 4. Analysis of Thematic Map of Author's Keywords.

3.5. Systematic Review

3.5.1. Fundamentals of PV-BESS Hybrid Systems

Hybrid systems that integrate photovoltaic (PV) solar generation with battery energy storage systems (BESS) represent a key solution to overcome the inherent challenges of solar energy, such as intermittency and variability [22,31]. These systems combine multiple energy sources to minimize the impact of seasonal fluctuations in solar irradiation and wind speed, thereby ensuring a more stable and reliable electricity supply [31]. The architecture of such systems is crucial for their efficiency and functionality, allowing them to operate either in standalone (off-grid) mode or connected to the main grid (on-grid) [32].

A typical PV-BESS hybrid system consists of several key components that work together. The photovoltaic (PV) panels are the primary generation component, converting solar radiation into direct current (DC) electrical energy [33,34]. Their performance is highly dependent on climatic factors such as irradiance and temperature [35]. The Battery Energy Storage System (BESS)—commonly based on lithium-ion or other advanced electrochemical technologies—stores excess energy generated by the PV panels for later use, for instance, during nighttime or cloudy days [36]. This component is essential for mitigating power fluctuations and ensuring continuity of supply [37].

To manage the energy flow between DC components and alternating current (AC) loads, converters and inverters are used. Inverters transform the DC output from panels or batteries into AC power to supply residential or commercial loads, while bidirectional converters manage the charging and discharging processes of the batteries [38].

System configuration can vary significantly. In off-grid systems, the BESS is indispensable to ensure a 24-h power supply [39]. In grid-connected systems, the BESS can optimize self-consumption, reduce electricity costs by storing energy when it is cheap and using it when prices are high (energy arbitrage), and provide backup power during grid outages [40]. The management and control of these systems are complex and often rely on an Energy Management System (EMS), which makes decisions on when to store, consume, or sell energy to the grid, prioritizing the use of renewable sources [41]. Optimal operation requires accurate prediction of the State of Charge (SOC) of the battery—a critical parameter to extend its lifespan and ensure safe and cost-effective use of energy storage [42]. The comparation of hybrid systems is shown in Table 3.

Table 3. Comparison of Hybrid System Components and Configurations.

Feature	Photovoltaic System	BESS (Battery Storage) System	Off-Grid Configuration	On-Grid Configuration
Main Function	Generating electricity from solar radiation [34,43].	Store excess energy for later use and mitigate fluctuations [42,44].	Supply energy autonomously without connection to the main power grid [41,43].	Complement the grid supply, optimize costs and sell surpluses [31,45].
Key Components	Photovoltaic modules (en. polySi), inverter, charge controller (MPPT) [46,47].	Batteries (e.g., Lithium-Ion), Battery Management System (BMS), Bi-directional Converter [48,49].	PV panels, BESS, backup generator (e.g., diesel or biogas), inverter for autonomous power without connection to the main power grid [41,42,50].	PV panels, BESS (optional but recommended), grid-connected inverter, bi-directional meter [22,41].
Dependency/Challenges	Highly dependent on weather conditions (irradiation, temperature, cloudiness) [22,31,43].	It depends on the energy generated by the PV system or the grid for charging. Its lifespan is affected by charge/discharge cycles [40,42].	Completely independent of the power grid. Reliability depends on system sizing and the availability of local resources [22,42].	Grid-dependent for backup and energy sales. Vulnerable to grid outages, although the BESS can provide autonomy [43,44].
Optimization Objective	Maximize energy generation through algorithms such as MPPT (Maximum Power Point Tracking) [31,43,51].	Optimize charge/discharge cycles to maximize service life and profitability, managing SOC and SOH [42,49].	Ensure reliability of supply (e.g., minimize LPSP—Probability of Loss of Power Supply) and autonomy [45,50].	Minimize energy costs (LCOE), maximize self-consumption and income from the sale of surpluses [22,31].
Energy Flow	One-way: from the sun to the System [45]. It is managed by EMS for efficient use [41].	Bidirectional: charging from PV/grid and discharging to loads/grid, managed by BMS and EMS [40,46,52].	The flow is managed internally to meet local demand. Excess flow is stored or discarded [41,42].	Bidirectional with the power grid: buying and selling energy [31,41].

3.5.2. Machine Learning Approaches for PV Generation Forecasting

Accurate forecasting of photovoltaic (PV) solar power generation is essential for the efficient management of microgrids, power system stability, and optimization of energy storage systems [53].The intermittent nature of solar energy, influenced by complex meteorological variables, often renders conventional forecasting methods insufficient [42].

For this reason, Machine Learning (ML) and Deep Learning (DL) techniques have become indispensable tools, owing to their ability to model complex nonlinear relationships and learn patterns from large volumes of historical and real-time data.

Forecasting approaches can be categorized according to the prediction horizon—ranging from very short-term (seconds to minutes), which is useful for real-time control and frequency regulation, to short-term (hours to days), medium-term, and long-term forecasts, applied in economic planning and plant maintenance [54].

A crucial step in developing robust forecasting models is data preprocessing and feature engineering [55]. The accuracy of any ML model largely depends on the quality and relevance of its input data. Meteorological variables such as solar irradiance, ambient temperature, cloud cover, humidity, and wind speed are among the most influential factors [34].

In fact, correlation analyses show that irradiance has the strongest positive correlation with PV power output, whereas temperature tends to have a negative impact on panel efficiency [31]. To enrich the dataset and improve model predictive capability, advanced feature engineering techniques are often employed. Among these, signal decomposition methods—such as the Wavelet Transform (WT) [42], Empirical Mode Decomposition (EMD) and its enhanced variants (EEMD, CEEMDAN, MEMD), and Variational Mode Decomposition (VMD) [56]—are widely used. These methods decompose PV power time series signals into multiple frequency components (Intrinsic Mode Functions, IMFs), enabling DL models to capture multi-scale patterns and reduce data non-stationarity. Additionally, clustering algorithms like K-means are applied to segment data according to weather conditions (e.g., sunny vs. cloudy days), allowing for the training of specialized models under each scenario [42]. The choice of ML or DL algorithm is critical for forecasting performance. The models can be grouped into several main categories:

Statistical and Regression Models

Statistical and regression models—including Linear Regression, Polynomial Regression, and the Autoregressive Integrated Moving Average (ARIMA) model—represent the traditional starting point for time series forecasting. Their main advantage lies in their simplicity and computational efficiency, as they do not require high processing power or long training times to produce predictions [34,41]. These models perform well when the relationship between input variables (e.g., historical generation data) and the output (future forecast) is relatively simple and follows linear or predictable patterns [47]. For example, the ARIMA model relies solely on historical data from the same time series to identify patterns and generate forecasts, serving as a useful benchmark against more advanced techniques [51]. However, their main weakness becomes evident when modeling PV generation, which is inherently complex and nonlinear. PV energy output depends on the dynamic interaction of multiple factors—mainly meteorological ones such as solar irradiance, temperature, humidity, and cloud cover [47]. These models struggle to capture nonlinear relationships because they assume a direct and proportional link between inputs and outputs. PV generation, however, does not behave in such a manner. For instance, solar panel efficiency decreases with increasing temperature—a nonlinear relationship that simple regression models fail to capture adequately [33].

Traditional models like ARIMA also assume stationary data, where statistical properties such as mean and variance remain constant over time. In contrast, solar generation is highly non-stationary, affected by diurnal and seasonal cycles and unpredictable weather fluctuations [55].

Although ARIMA employs a differencing process to stabilize the series, it is often insufficient to handle the high volatility of PV data. Moreover, traditional models are typically univariate, relying only on past values of the same variable and unable to natively incorporate exogenous variables such as temperature or irradiance forecasts—key drivers of PV generation [54].

Although variants like ARIMAX include these variables, the model structure remains fundamentally linear. ML and DL models, by contrast, can integrate multiple meteorological and contextual variables to enhance predictive performance [53]. Statistical and regression models also struggle with abrupt changes and high variability; they perform better in predicting smooth trends but fail to handle sudden fluctuations caused by transient meteorological events (e.g., passing clouds) [31]. This intermittency is a defining characteristic of solar power, and DL architectures such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks are far better equipped to learn from these dynamic and volatile patterns [56].

Classical Machine Learning Models

Classical Machine Learning (ML) models are fundamental tools for optimization and prediction in energy systems, particularly within the context of microgrids and renewable energy integration [57]. Various studies have employed algorithms such as Linear Regression (LR), Support Vector Machines (SVM), Artificial Neural Networks (ANN), Random Forest (RF), and Decision Trees for forecasting and classification tasks [55].For instance, in the prediction of photovoltaic (PV) solar energy generation, nine different models were systematically compared, concluding that ANN was the most effective, achieving a Root Mean Square Error (RMSE) of 274.84 kWh and a Mean Absolute Percentage Error (MAPE) of 5.26% [54]. Similarly, another study forecasting PV production and residential load demand found that the Random Forest model provided the highest accuracy, with R^2 values of 0.92 for generation and 0.90 for demand. These findings demonstrate that, although multiple viable models exist, the selection of the optimal algorithm depends on the specific dataset and the problem being addressed [58].

Artificial Intelligence (AI) and classical ML models are also crucial for the efficient management of Battery Energy Storage Systems (BESS), particularly in Electric Vehicles (EVs). Algorithms such as Feedforward Neural Networks (FFNN), SVM, Random Forest, and Fuzzy Logic (FL) are employed to estimate the State of Charge (SOC) and State of Health (SOH) of batteries [49]. The main advantage of these models lies in their ability to handle complex data and nonlinear relationships. For example, SVMs are effective for SOC estimation, offering high accuracy and avoiding overfitting, although their computational complexity and processing time can be significant [42]. On the other hand, FFNNs are easier to implement but may require long training times and risk converging to local minima. AI has also been applied to parameter optimization in controllers, such as the Levenberg–Marquardt (LM) algorithm used in Home Energy Management Systems (HEMS), which has demonstrated superior performance compared to other methods like Bayesian Regularization (BR) and Scaled Conjugate Gradient (SCG) [59].

In energy applications, ML algorithms are typically categorized into supervised, unsupervised, and reinforcement learning approaches [33]: Supervised learning, including ANN, SVM, KNN, Random Forest, and Decision Trees, is the most widely used for forecasting and classification tasks [60]; Unsupervised learning is applied to clustering through methods such as k-means or Gaussian mixture models; Reinforcement learning addresses sequential decision-making for dynamic energy management and the control of complex systems [33].

In parallel, Deep Learning (DL) models—such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), including LSTM and GRU variants—have established a new paradigm capable of processing raw data and learning hierarchical representations without manual feature extraction [60]. Nevertheless, DL adoption faces several challenges: (i) the need for large volumes of high-quality data, often limited in new installations or in regions with scarce monitoring infrastructure [56]; (ii) the dependence on a selection of variables and rigorous preprocessing; and (iii) the interpretability of black box models, critical for reliability in control and operation. (iv) dependence on rigorous variable selection and preprocessing; and (v) the "black-box" nature of models, which limits interpretability—an essential aspect for reliability in control and operational applications [55].

Artificial Neural Networks (ANN)

Artificial Neural Networks (ANN) are a fundamental technique within Machine Learning, inspired by the structure and function of biological neural networks, and have demonstrated remarkable effectiveness in forecasting renewable energy systems [61]. Their

architecture consists of interconnected processing units, known as neurons or nodes, organized into layers (input, hidden, and output), which process information and make decisions by adjusting the weights of their connections during a training phase [42].

This capability enables them to learn complex and nonlinear relationships between multiple input variables (such as irradiance, temperature, humidity, and historical data) and photovoltaic (PV) power generation—a task where simple linear regression models often fail [45]. ANNs are particularly valuable in energy forecasting applications because they can be trained to map the functional relationship between variables from a known set of representative values, allowing them to generalize results to unseen data [62].

In the context of energy systems, ANNs are widely used for electric load forecasting, solar energy generation prediction, electric vehicle demand estimation, and energy management optimization in microgrids, demonstrating their versatility [22]. In fact, comparative studies have shown that well-tuned ANN models can outperform other ML models such as SVM or Decision Trees, achieving high accuracy even with limited datasets [31,54]. However, ANN performance critically depends on proper training, which is generally carried out using algorithms such as Levenberg–Marquardt or Backpropagation. These methods, while effective, may suffer from disadvantages such as slow convergence or entrapment in local minima, which has motivated the adoption of evolutionary algorithms to optimize weights and improve overall performance [63]. Despite their effectiveness, ANNs also present challenges, such as the risk of overfitting, the need for large amounts of high-quality training data, and their "black-box" nature, which reduces interpretability compared to simpler models [40].

Deep Learning Models for Time Series

Deep Learning (DL) models are advanced computational tools that have proven highly effective for time series forecasting in the energy sector, particularly for predicting photovoltaic (PV) power generation and load demand [61]. These models, which fall under the broader domain of Machine Learning (ML) and are based on Artificial Neural Networks (ANN), can process raw data without requiring manual feature extraction, distinguishing them from traditional ML models [60]. Their main advantage lies in their ability to handle large datasets and capture complex nonlinear relationships and temporal dependencies [56], which is crucial for forecasting the inherent variability of renewable sources such as solar energy [64]. Algorithms such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) are particularly suitable for sequential data, while Convolutional Neural Networks (CNN) excel at extracting spatial features from data [60].

Numerous studies have compared different DL architectures to determine which offers the greatest forecasting accuracy. Hybrid models, which combine the strengths of various architectures, often outperform single models [58]. For example, a hybrid CNN–LSTM model uses CNNs for pattern extraction and LSTMs for learning temporal dependencies, achieving a lower Root Mean Square Error (RMSE) than standalone LSTM or Bi-LSTM models [55]. Similarly, Transformer Networks—an emerging architecture that employs self-attention mechanisms—have demonstrated superiority over RNNs by overcoming issues such as gradient vanishing and by being able to analyze temporal patterns across long sequences [65]. A comparative study found that a Transformer model (PVTrans-Net-EDR) integrating a pre-trained LSTM network to enhance meteorological forecasting reduced the Mean Absolute Error (MAE) by up to 56.9% compared to a simple LSTM model. Another study concluded that a hybrid CARIMA–SARIMA–LG model outperformed MLP, RBF, and ensemble models (Bagging and Boosting) in predicting the active power of solar tracking systems [47].

The effectiveness of DL models largely depends on the quality and preparation of input data. Feature engineering is a crucial step that involves selecting, extracting, and generating relevant variables to improve the model's predictive capability [54]. Signal decomposition techniques such as Wavelet Transform (WT), Empirical Mode Decomposition (EMD), and Variational Mode Decomposition (VMD) are used to break down time series into simpler and more stable components, facilitating the model's ability to learn underlying patterns [56]. For instance, one study proposed a modified EMD (MEMD) that generated features more strongly correlated with solar power generation, thereby improving forecasting accuracy. Additionally, clustering algorithms such as K-means are employed to group data with similar characteristics, enhancing the model's adaptability to different conditions [55]. Data preprocessing, including normalization and the handling of outliers and missing values, is also fundamental for ensuring robust model performance [62]. The different applications of ML and DL for battery State of Charge (SOC) prediction, hybrid system optimization, and power quality control in Table 4.

Table 4. Application of Machine Learning (ML) and Deep Learning (DL) for battery State of Charge (SOC) prediction, hybrid system optimization, and power quality control.

R	Objective	Methodology	Results	Context	Limitations
[42]	Predicting the State of Charge (SOC) of lead-acid batteries using neural networks for a solar drying system (Solar Smart Dome 4.0).	- Models: Transformer Neural Network, LSTM, GRU. Input data: Battery voltage, current and temperature.	- GRU was the most accurate model: MAE 0.642%, RMSE 0.885%, R^2 99.88%. The Transformer model was faster to train but less accurate than LSTM and GRU.	Agricultural drying system in rural areas, powered by photovoltaic solar energy with battery storage.	It demonstrates the generalization capabilities of neural networks to predict SOC across different battery types and with limited data, which is vital for storage management.
[31]	Design and optimize a grid-connected residential microgrid with PV, wind turbines, and BESS, using ML to predict total renewable generation.	- Software: HOMER Pro version 3.16.2 for techno-economic optimization. ML model: Random Forest Regressor for forecasting. Input data: Irradiance, temperature, wind speed.	- Random Forest had high accuracy: R^2 of 0.99996, MAE of 0.0122, RMSE of 0.0287 for total renewable power prediction. The optimal configuration (PV + WT + BESS + Grid) achieved a renewable fraction of 68.3% and a COE of $0.035/kWh.	Residential microgrid in Pabna, Bangladesh, to mitigate the impact of grid outages and reduce energy costs and emissions.	Integrates ML prediction directly into the design and optimization of a hybrid microgrid, validating its stability and economic viability in a real-world context with grid problems.
[66]	Develop a two-stage energy management framework for a hydrogen-power distribution network with PV-enabled hydrogen fueling stations (HFS).	- Stage 1: MISOCP optimization for daily scheduling of HFS, PV, OLTC and capacitor banks. - Stage 2: DRL (SAC algorithm) to reschedule the reactive power of PV systems and the power/hydrogen of HFS in real time.	- The two-stage frame was effective in maintaining stable operation of the PHCIES and maximizing HFS gains. The adaptive safety module in the DRL eliminated SOH and voltage violations during training.	Power-Hydrogen Integrated Energy System (PHCIES) with stand-alone PV systems and PV-enabled Hydrogen Fueling Stations (HFS).	It proposes a hybrid approach that combines classical optimization and secure DRL for the complex management of multi-energy systems (electricity and hydrogen), ensuring safe and profitable operations.
[63]	Designing an AI controller for a UPQC integrated with a solar-battery system in three-phase distribution networks to improve power quality.	- AI Models: ANN trained with Soccer League algorithm (S-ANNC) for the shunt active power filter. FLC for the series active power filter. Synchronization Technique: STF-UVGM to eliminate the need for PLL.	- The S-ANNC controller outperformed GA, PSO, and GWO methods in decreasing THD, improving power factor, and mitigating voltage distortions. The system maintained constant DC link voltage during load/irradiation variations.	UPQC in three-phase distribution networks with PV and battery storage integration to mitigate power quality (PQ) problems.	It demonstrates the superiority of a hybrid AI controller (S-ANNC and FLC) for managing UPQC in systems with PV and storage, improving power quality without the need for traditional components such as PLLs.
Touzani [67]	Controlling distributed energy resources (DERs) through DRLs for load flexibility and energy efficiency in buildings.	- DRL Model: Deep Deterministic Policy Gradient (DDPG) algorithm for the integrated control of HVAC systems and electric battery storage with on-site PV generation.	- The DRL-based controller achieved cost savings of up to 39.6% compared to a rule-based baseline controller while maintaining similar thermal comfort.	Integrated control of HVAC systems and electric battery storage in the presence of on-site PV generation in commercial buildings.	Contribution: Implements and validates a DRL controller in a physical building for integrated DER management, demonstrating its effectiveness in reducing energy costs and managing load flexibility in a real-world environment.

Note. R: Reference.

3.5.3. ML/DL Models in Prediction, Grid Operation, Energy Control, and Battery Management with AI

Hybrid models that combine Machine Learning (ML) and Deep Learning (DL) techniques are increasingly used to enhance the accuracy of time series forecasting, such as energy generation and load demand prediction [65]. A common approach involves employing ML algorithms—such as Principal Component Analysis (PCA) or clustering methods (K-means)—for feature engineering and selection, and then feeding the processed data into DL models like Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), or Gated Recurrent Units (GRU), which are effective at capturing temporal dependencies [55]. For instance, one study proposed a hybrid model that uses a clustering algorithm (K-means) to group data and then a CNN network for hourly load forecasting [60]. Another successful approach is the hybridization of different forecasting models, such as CARIMA–SARIMA–LG, which integrates a Controlled Auto-Regressive Integrated Moving Average model (CARIMA), a seasonal model (SARIMA), and a logistic distribution (LG) to achieve high-accuracy forecasts for solar tracking photovoltaic (PV) systems, reaching an R^2 of up to 0.983 [47]. Combining these models enables the handling of both seasonal patterns and complex nonlinearities inherent in energy data.

Hybrid models are essential for the efficient operation and control of microgrids and electric vehicle (EV) systems [34]. In energy management, control systems have been developed that integrate classical optimization algorithms (such as Mixed-Integer Linear Programming—MILP) with DL models to support real-time decision-making [68]. One example is an energy management controller for EV charging stations that employs an Artificial Neural Network (ANN) trained with data optimized by the Soccer League Algorithm (S-ANNC) to manage power flow and maintain voltage stability in the DC link [63]. Similarly, systems combining PID controllers with swarm optimization algorithms (such as Particle Swarm Optimization, PSO) and neural networks (ANN) have been proposed for Maximum Power Point Tracking (MPPT) in PV systems [33]. These hybrid approaches enable rapid and adaptive responses to changing grid conditions, improving stability, reducing costs, and optimizing the use of distributed energy resources.

Beyond prediction and control, hybrid ML/DL models are powerful tools for fault diagnosis and stability assessment in energy systems [66]. For example, in fault detection for sensors in Battery Energy Storage Systems (BESS), a hybrid model has been proposed that uses PCA for anomaly detection and spectral residual (SR) analysis to identify the faulty sensor [69]. n grid stability assessment, methods combining CNNs with transfer learning have been developed to evaluate voltage instability in DC microgrids. Specifically, a pre-trained model such as ResNet-50 is used, with its learned features transferred to classify instability events, achieving high accuracy (99.8%) with a relatively small dataset [70]. These hybrid models outperform conventional methods due to their robustness, scalability, and reduced need for large labeled datasets for training.

The main advantage of hybrid ML/DL models lies in their ability to combine the strengths of different algorithms to tackle complex problems more effectively than single models [57]. For instance, integrating signal decomposition techniques (such as Wavelet Transform or Empirical Mode Decomposition, EMD) with LSTM networks allows for better feature extraction and more accurate PV generation forecasting [55]. Likewise, the integration of optimization algorithms (such as PSO or Genetic Algorithms) with neural networks enables optimal tuning of controller parameters to improve energy management and battery lifespan. However, these approaches are not without challenges [71]. Computational complexity remains a major concern, as combining multiple models can demand significant resources and training time [67]. Moreover, the interpretability of hybrid DL models—often regarded as "black boxes"—can hinder understanding of their decision-making processes,

which is crucial for ensuring reliability in critical power grid applications [60]. Different studies about forecasting are shown in Tables 5 and 6.

3.6. Meta-Analysis

A random-effects meta-analysis (REML estimator) was conducted on k = 5 studies reporting R^2 values of ML/DL models for photovoltaic and/or demand forecasting: Manlapaz [53], Mollik [58] (two estimates: generation and demand), Habib & Hossain [55], and Ali [31]. The analysis was performed using the R^2 values as reported in the original articles (where applicable) and without reported sample variances; therefore, external weights were applied (Figure 5).

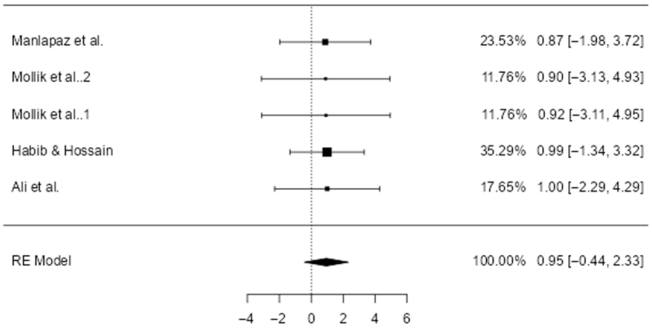


Figure 5. Forest plot of the meta-analysis (random effects) of R^2 in PV/load forecast (k = 5). Manlapaz [53], Mollik [58] (for both mentions), Habib & Hossain [55], Ali [31].

3.6.1. Weights for Analysis

Hybrid models that combine Machine Learning (ML) and Deep Learning (DL) techniques are used to improve the accuracy of time series forecasting, such as energy generation and load demand prediction [65]. A common approach involves employing ML algorithms—such as Principal Component Analysis (PCA) or clustering methods (K-means)—for feature engineering and selection, and then feeding these processed data into DL models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), or Gated Recurrent Units (GRU), which are effective at capturing temporal dependencies [55].

3.6.2. Combined Effect

The combined mean was $R^2 = 0.95$ with a 95% CI [−0.44; 2.33], Z = 1.13, $p = 0.260$. Although the *p*-value does not reach significance at the 95% level, the point estimate suggests that, on average, the models explain 95% of the variability of the series (PV/load forecasting) in the studied contexts. The lack of statistical significance is explained by the low precision (wide confidence interval) resulting from the absence of reported variances or standard deviations for each study; under this scenario, the interval is constructed under normality assumptions and may exceed the logical limits of R^2.

3.6.3. Heterogeneity

The heterogeneity statistics indicate $I^2 = 0\%$, $\tau^2 = 0$, Q = 0.004 ($p = 1.000$). This means that no variability was detected among studies beyond sampling error. Methodologically, this is consistent with: (i) a small set of *k* studies, (ii) relatively similar R^2 results (0.87–1.00), and (iii) the use of approximate weights (not derived from variances). In practical terms, this suggests consistency in the performance of the models across different contexts (buildings/microgrids, residential PV–BESS, campus systems), in line with what was reported in the individual studies (e.g., high R^2 values in [31,53,58]) (Figure 6).

Table 5. Research that develop and compare ML/DL models specifically for forecasting solar photovoltaic generation.

Reference	Main Objective	Methodology and Models Used	Key Results and Performance Metrics	Input Data and Preprocessing	Limitations and/or Contributions
[34]	Propose a long-term ML-based solar PV generation forecasting framework for effective operation of system operators.	- Model: ML framework based on regression analysis, comparing a standard base model with a proposed auxiliary cascade model.	- The proposed framework improves the accuracy of the Renewable Energy Forecast (REF) by incorporating observational variables along with the forecasting model.	- Input Data: GHI, DNI, ambient temperature, humidity, cloud variation, seasonal variation.	Contribution: The framework is vital for long-term forecast horizons, enabling better estimation of solar PV potential, which is crucial for planning large-scale storage systems.
[55]	Introduce a framework for forecasting short-term PV generation in isolated microgrids, based solely on solar irradiance data from remote weather stations.	- Hybrid Model: Two-stage Hybrid Data Linked Model (HDLM) architecture, which integrates a Layered Recurrent Neural Framework (LRNF) and a pattern identification network. Feature Engineering: Modified EMD (MEMD) and K-Means clustering.	- HDLM outperformed other models: MAE of 1.02, RMSE of 2.176, and R^2 of 0.991. Computation time was reduced by 15% after feature selection.	- Input Data: Only solar irradiance data from a distant weather station. Preprocessing: MEMD for signal decomposition and feature generation.	Contribution: Demonstrates that highly accurate PV forecasting can be achieved for remote locations (microgrids) even with limited data (irradiance only), thanks to advanced feature engineering.
[56]	To conduct a systematic literature review on Deep Learning applications for solar PV forecasting.	- Methodology: Systematic review of 26 selected articles, analyzing DL architectures, preprocessing techniques, input features and evaluation metrics.	- LSTM was the most used algorithm (32.69%), followed by CNN (28.85%). Wavelet Transform (WT) was the most prominent data decomposition technique, and Pearson Correlation was the most used for feature selection.	- Most Common Input Features: Ambient temperature, pressure and humidity. Preprocessing Techniques: WT, VMD, K-Means, GANs.	Contribution: Provides a structured overview of the state of the art in solar forecasting with DL, identifying the most effective models and techniques and the persistent challenges such as model generalization and interpretability.
[61]	Investigate the benefits of combining an ML approach with PV energy forecasts generated by meteorological models.	- ML Models: Linear Model, LSTM, XGBoost, LightGBM. - Combined Approach: Uses forecasts from a numerical weather model (NWP) as input for the ML models.	- Linear models were the most effective, with an RMSE improvement of at least 3.7% in PV production forecasting compared to the reference methods.	- Input Data: PV production forecasts from reference models (BaselineP and BaselineD) and past production observations.	Contribution: Demonstrates that simple ML models, such as linear ones, can significantly refine and improve the forecasts of established physical models, even with limited data, which is useful for newly installed plants.
Kim [65]	Develop Transformer network variants (PVTransNet) for next-day hourly PV power forecasting.	- Models: Three Transformer variants (PVTransNet-E, PVTransNet-ED, PVTransNet-EDR) and LSTM base models. - PVTransNet-EDR combines LSTMs to improve input weather forecasts.	- PVTransNet-EDR outperformed all models: It reduced the MAE by up to 56.9% and improved the R^2 by 0.7062 compared to a simple LSTM model.	- Input Data: Historical PV power generation, meteorological observations, weather forecasts, and solar geometry data.	Contribution: Introduces a specialized Transformer architecture that integrates a pre-trained LSTM model to improve the accuracy of PV generation forecasting, demonstrating the superiority of attention mechanisms for this task.
Khadeeja [54]	Predicting daily solar power generation using ML, with a focus on application to a newly installed solar plant with limited data.	- ML Models: Linear Regression, ARIMA, ANN, SVM, Random Forest, Decision Tree, GBM, LGBM, XGBM.	- ANN was the most effective model, achieving an RMSE of 274.84 kWh, MAE of 245.93 kWh and a MAPE of 5.26%.	- Input Data: Irradiance, humidity, minimum and maximum temperatures, and surface pressure.	Key Contribution: Demonstrates the feasibility of achieving accurate forecasts with a limited data set from a newly installed solar plant, a practical and underexplored scenario in the literature.

Table 6. Studies using AI for battery management in various applications, including electric vehicle (EV) charging.

Reference	Objective	Methodology	Results	Context	Limitations
[40]	Determine the optimal sizing of deep-cycle batteries and manage their charging in real time in Solar-EV systems.	- Simulation Model: MATLAB/Simulink R2021a (version 9.10) with RTTM for real-time analysis. ML Models: Gradient Boosting, Random Forest Regression, ANN, SVM, Decision Tree, MLP, etc., to predict the DoD.	- Random Forest was the best DoD predictor, with an accuracy of 99.998%. Integrating MPPT and a PLL controller improved system efficiency and grid synchronization.	Charging electric vehicles (EVs) using a hybrid solar-grid system based on UDDS driving cycles.	Contribution: Provides an integrated framework for intelligent battery sizing and management for EVs, validated with a real-time hardware platform (RTTM) and optimized through a comprehensive comparison of ML models.
[49]	Provide a comprehensive analysis of intelligent control strategies and BMS methodologies in EV applications.	- Methodology: Literature review on optimization algorithms (GA, PSO) and ML (FFNN, SVM, RF, FL) for estimating the battery state (SOC, SOH).	- ML and DL algorithms have shown superior results in battery health estimation, although they require large, high-quality data sets and fast processors.	Battery management system (BMS) in electric vehicles (EVs), covering state estimation, cell balancing, fault diagnosis, and thermal control.	Contribution: Provides a comprehensive and critical review that classifies and evaluates various AI techniques for BMS, identifying their advantages, disadvantages, and implementation challenges in the context of EVs.
[72]	Investigate the role of AI in developing smart EV charging infrastructure in Malaysia.	- Methodology: A mathematical model was developed for an AI-based smart charging system. Technologies such as ML and predictive analytics for charging management were reviewed.	- The implemented AI-based smart charging system achieved 30% energy savings and 20.38% cost reduction compared to traditional methods.	Electric vehicle charging infrastructure (EVCS) in Malaysia, with a focus on smart charging, demand management, and renewable energy integration.	Contribution: Demonstrates, using a mathematical model and cost analysis, the quantitative benefits of AI for optimizing EV charging, improving efficiency, and supporting grid stability in a specific national context.
[48]	Develop an AI-based energy management controller for a DC microgrid-based EV charging station, with V2G and G2V capabilities.	- AI Model: ANN with an adaptive interaction algorithm for energy management. MPPT Control: ANFIS to maximize the power of the PV system.	- The ANN-based PMC controller reduced the DC bus voltage overshoot from 9.6% to 0%, the stabilization time from 1.18 s to 0.52 s, and the rise time from 0.27 s to 0.25 s, compared to a conventional controller.	Electric vehicle charging station (EVCS) based on a DC microgrid with a PV system, storage battery, and grid connection. It operates in V2G and G2V modes.	Contribution: Proposes and validates an adaptive ANN controller for bidirectional power management in an EVCS, significantly improving DC bus stability and power management efficiency.
[73]	Developing an off-grid PV-based hydrogen and oxygen co-production system using ML and multi-objective optimization.	- ML models: Gaussian Process Regression (GPR) and Weighted Average Surrogate (WAS) to establish a surrogate model between the design variables and the optimization objectives. - Optimization algorithms: POGCEA and RVGEA.	- The optimized system increased hydrogen production by 16.80% and PEM electrolysis efficiency by 12.08% compared to the initial solution.	Off-grid hydrogen and oxygen co-production system consisting of PV-BESS-PEM for continuous and stable hydrogen production around the clock.	Contribution: Proposes a methodological framework that integrates ML models and multi-objective optimization algorithms for the design and operation of green hydrogen production systems, maximizing both production and efficiency.

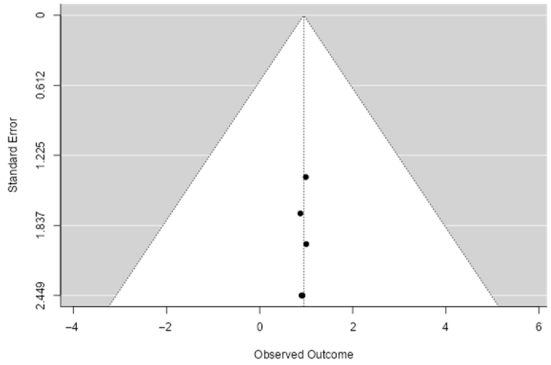


Figure 6. Funnel chart for R^2 in PV/load forecast [31,53,58].

3.6.4. Publication Bias

All three tests were non-significant: Egger ($p = 0.980$), Kendall's Tau ($p = 0.448$), and Fail-Safe N (Rosenthal) = 0, $p = 0.137$. The funnel plot appears symmetrical. With a small number of studies ($k = 5$), these tests have low statistical power, but there is no evidence of publication bias.

3.6.5. Equivalence Tests (TOST)

The lower test is significant ($p = 0.043$), while the upper test is not ($p = 0.702$). Under the default equivalence bounds, this suggests that the combined effect exceeds a lower threshold (i.e., it is not "too low"), although it cannot be confirmed to lie entirely within a narrowly predefined margin. Since R^2 is a bounded metric within [0, 1], the interpretation here is more descriptive than inferential.

4. Discussion

4.1. Superiority of ML/DL Models in PV

The results of the systematic review confirm that machine learning (ML) and deep learning (DL)-based approaches consistently outperform traditional forecasting methods (ARIMA, regression models, persistence, etc.) in highly variable photovoltaic (PV) scenarios. In particular, sequential models such as LSTM networks and hybrid CNN-LSTM architectures more effectively capture the nonlinear and stochastic nature of solar generation than classical techniques, significantly reducing prediction error. Consistently, the quantitative meta-analysis conducted in this study showed that, on average, ML models explain 95% of the variability in PV generation, quantitatively corroborating their high predictive effectiveness. Although this combined estimate did not reach statistical significance due to the limited number of studies, no heterogeneity was detected among them, suggesting general consistency in the benefits of applying ML across different contexts. Likewise, advanced architectures with Transformer-type attention mechanisms are emerging with outstanding results: some next-generation Transformer variants have outperformed both climatological persistence and conventional recurrent neural networks in intraday PV power forecasting.

Indeed, hybrid combinations integrating CNN, LSTM, and Transformer networks have reported the highest accuracy compared to existing models, demonstrating the advantage of extracting spatial features with CNNs and temporal dependencies with memory/attention mechanisms. These findings highlight that deep learning techniques—particularly hybrid CNN-LSTM and Transformer models—offer greater accuracy and capacity to generalize complex patterns, especially under highly fluctuating meteorological conditions. However, it should be noted that most Transformer-based forecasting studies to date have focused on pure PV power prediction alone, without directly evaluating whether such

accuracy improvements translate into enhanced operational performance of the energy storage system.

Practically, these improvements in predictive capacity translate into more efficient operation of PV-ESS: more accurate forecasts enable optimized battery charge/discharge cycles, prevent overloading, and enhance grid stability. Academically, these results consolidate the evidence supporting deep learning approaches in the photovoltaic domain, providing an integrative synthesis that had not previously existed to comparatively evaluate different PV-ESS forecasting techniques from a unified perspective.

4.2. Future Directions for Improving the Forecast

In light of these limitations, several future research directions are proposed to strengthen the application of AI in PV-ESS forecasting. First, it is essential to develop standardized benchmarking frameworks for evaluating ML models in this domain. This involves defining reference datasets, common forecasting horizons, and unified metrics so that different techniques can be compared under equivalent conditions. The adoption of consistent benchmarks would allow a more objective determination of which algorithms provide advantages under specific climatic or operational scenarios. Second, it is proposed to complement traditional accuracy metrics (RMSE, MAE, MAPE, etc.) with indicators of forecast robustness and resilience—that is, to evaluate how model performance degrades under atypical events, missing data, or high intermittency, as well as to assess the impact of forecasting errors on the stability of microgrids with limited resources. Incorporating resilience metrics would provide a more practical perspective on model reliability beyond average accuracy, which is particularly relevant in isolated or low-flexibility operational environments.

4.3. Adaptive Control and Integrated Operation

The integration of Deep Reinforcement Learning (DRL) techniques emerges as a promising path for adaptive control of charging and discharging in PV-ESS. Unlike static control schemes, a DRL-based agent can learn optimal battery management policies in real time from environmental feedback, dynamically adapting to changing conditions. Recent literature already highlights the relevance of intelligent, AI-based control strategies for optimizing renewable microgrids; extending these approaches through DRL would enable the closure of the forecast–action loop, where solar predictions directly feed autonomous and optimal ESS operational decisions. In parallel, to address the scarcity of historical data in sites with limited instrumentation, the generation of synthetic or augmented data becomes essential. Modern generative methods, such as Generative Adversarial Networks (GANs), can create artificial meteorological and solar production data, reproducing realistic patterns that expand available training datasets. Several studies report that the use of synthetic data improves the robustness and generalization capacity of models in data-limited environments by mitigating small-sample bias.

4.4. Interpretability and Computational Efficiency

The adoption of Explainable Artificial Intelligence (XAI) techniques is emphasized to improve the interpretability of deep learning models in energy applications. Currently, algorithms such as LSTM or Transformers operate as "black boxes," difficult to interpret compared to statistical models with transparent parameters. The application of XAI—such as variable importance attribution, visualization of attention maps, or SHAP value analysis—would help elucidate the internal logic of these networks, fostering trust among operators and planners in the model's recommendations. This line of research is crucial for aligning AI solutions with the practical challenges of energy planning in regions with

low grid reliability and high solar intermittency, where understanding why a model makes certain decisions is as important as the accuracy of its forecasts.

4.5. Extension to Other Technologies

It is worth noting that many of these perspectives can be extended to other emerging energy technologies beyond photovoltaics. For example, in green hydrogen production via electrolysis, having accurate forecasts of renewable generation (solar and wind) and energy demand would optimize electrolyzer operation and hydrogen storage management, increasing system efficiency. Similarly, in smart grids, AI-based forecasting algorithms can anticipate fluctuations in both distributed generation and consumption, facilitating the optimal coordination of distributed resources (photovoltaic, wind, electric vehicles, etc.) and improving grid stability. In the case of solar thermal energy, advanced predictive models of solar resources combined with thermal storage strategies could maximize usable heat production and supply reliability. Overall, extending these analytical techniques to emerging domains would broaden the impact of artificial intelligence in the energy transition by providing forecasting and management solutions applicable across multiple renewable technologies and storage systems.

Over multi-year horizons, PV generation and ESS behavior are inherently non-stationary due to factors like panel degradation, battery aging, climate variability, and shifting load patterns. To maintain forecasting accuracy as these conditions evolve, adaptive learning strategies are essential. One approach is online (incremental) learning, where the model is continuously updated with new data to capture changing patterns—such continuous model tuning has been shown to effectively adapt to concept drift and emerging trends in energy data [74]. Another strategy is transfer learning, which leverages knowledge from previously trained models to quickly recalibrate forecasts under new conditions [75]. Finally, continual (lifelong) learning frameworks offer a long-term solution by incrementally learning from data streams over time while avoiding catastrophic forgetting, thus balancing adaptability with stability [76]. These adaptive techniques ensure that the forecasting model remains robust and reliable over multiple years, preserving performance stability even as system characteristics and environmental conditions gradually change.

Recent studies highlight that BESS controllers must consider changes in battery characteristics due to aging (e.g., the progressive reduction in available peak power) to protect the system and extend its lifespan [77]. In fact, closed-loop approaches incorporating battery condition feedback have already been proposed: for example, adapting the charge/discharge protocol in real time according to the state of health (SoH) using optimal control, allowing for the detection of incipient failures and mitigating their impact by adjusting operation on the fly [78]. Including a discussion of these adaptive architectures would give the reader insight into how to address degradation with models that learn from the current state of the battery, thereby extending the lifespan of the energy storage system [79].

As ML models grow more complex, their computational and energy requirements surge, potentially undermining the intended environmental benefits of their applications. In fact, the computational effort for cutting-edge deep learning exploded by roughly 300,000× between 2012 and 2018, with a commensurately large carbon footprint. This reality has prompted calls for "Green AI" approaches that treat efficiency (energy or compute cost) as a key evaluation metric alongside accuracy [80]. In practice—especially for off-grid or resource-constrained settings like embedded PV–ESS forecasting—this means carefully balancing model accuracy against training and inference energy cost. One promising strategy is to adopt lightweight architectures that offer a favorable accuracy–energy trade-off. Techniques such as model quantization and pruning can dramatically shrink

model size and computational overhead, drastically cutting energy consumption while sacrificing only minimal accuracy [81]. Indeed, studies show that combining quantization-aware training with structured pruning yields significant energy savings for negligible drops in accuracy. By deploying suitably optimized small models on microcontroller-class edge devices (TinyML), one can perform PV–ESS forecasting locally with very low power usage–provided the models remain sufficiently accurate yet simple enough to run on such limited hardware [82]. This emphasis on energy-efficient model design addresses the reviewer's concern by ensuring that improvements in accuracy do not come at an untenable computational energy cost, thereby preserving the net environmental benefits of the solution.

4.6. Operational and Economic Implications of Forecasting Errors in PV–ESS

Although machine learning (ML) and deep learning (DL) models have significantly improved photovoltaic (PV) generation forecasting, the residual forecasting errors that persist in practical operation still have critical implications for the performance, reliability, and economics of photovoltaic–energy storage systems (PV–ESS). Imperfect forecasts propagate through the control layer and affect decisions related to battery charge and discharge scheduling, leading to multiple undesired operational outcomes.

Battery over-cycling is one of the most direct consequences of forecast inaccuracies. When generation is overestimated, the control system may discharge the battery prematurely in anticipation of solar input that fails to occur, forcing a subsequent unplanned recharge from the grid. Similarly, underestimating irradiance can delay charging, leading to missed opportunities for low-cost energy storage. Both situations result in additional charge/discharge cycles—often partial and closely spaced—that accelerate cell wear and reduce usable lifetime. Empirical analyses demonstrate that excess cycling and irregular depth-of-discharge (DoD) events substantially increase the capacity fade rate and thermal stress in lithium-ion batteries, shortening service life and increasing replacement frequency [83]. For example, improved forecast accuracy in PV capacity firming operations has been shown to reduce battery throughput by up to 29% and high-depth discharge cycles by 51%, directly mitigating degradation [84].

Beyond the technical implications, forecasting errors also degrade economic performance. Inaccurate predictions distort energy-arbitrage strategies, leading to suboptimal charging or discharging decisions relative to electricity price signals. Missing high-price discharge windows or charging during unfavorable periods can significantly erode profitability [85]. When coupled with accelerated degradation, these effects increase operational costs and shorten the economic payback period of the storage system. Recent optimization studies demonstrate that incorporating aging-aware cost functions and forecast uncertainty into scheduling algorithms can increase revenue per unit of capacity loss by over 20% compared with conventional control approaches [86].

At the system level, forecast-induced inefficiencies diminish the long-term return on investment (ROI) by compounding three effects: (1) accelerated battery wear and earlier replacement costs, (2) missed arbitrage revenue due to poor scheduling, and (3) increased reliance on the grid during forecasting errors. Together, these factors may offset the economic benefits that motivate ESS adoption in PV systems.

4.7. Forecasting Horizons and Their Operational Relevance in PV–ESS

An essential aspect of integrating photovoltaic (PV) forecasting with energy storage systems (ESS) is understanding how different forecasting horizons align with specific operational functions in real-world grid management. Forecasts at sub-minute to minute scales (seconds to tens of seconds) are crucial for frequency regulation and primary control

services, enabling the ESS to inject or absorb power instantaneously and maintain system stability under transient fluctuations [87]. The very short-term horizon (5–15 min) supports ramp-rate control and power quality by allowing the ESS to anticipate and smooth rapid PV output drops caused by passing clouds, preventing steep power gradients that could destabilize the grid [88]. Moving toward intra-hour and short-term forecasts (1–6 h), these are fundamental for intra-day operational scheduling, guiding charge–discharge cycles to optimize solar energy utilization and minimize grid dependence in microgrids or distributed energy systems [89]. Finally, day-ahead forecasts (24 h) inform strategic planning and economic dispatch, allowing operators to align ESS operation with expected demand peaks, price signals, and market participation opportunities [90]. This taxonomy from second-level predictive control to 24-h operational scheduling provides a clear framework linking forecast temporal resolution with ESS use cases, highlighting that forecasting precision at each horizon directly determines how effectively storage can support grid reliability, economic optimization, and renewable integration.

4.8. Challenges in Integrating Machine Learning-Based Forecasting with Energy Storage System Control

The intersection between machine learning (ML)-based forecasting and energy storage system (ESS) control remains relatively underdeveloped despite rapid advances in both domains, primarily due to several technical and structural challenges. First, there is a persistent data availability gap: high-resolution, synchronized datasets combining PV generation, meteorological variables, and detailed ESS operation parameters are scarce. Many ML forecasting studies rely on isolated PV datasets without battery operation data, limiting the direct applicability of predictive models for integrated control tasks [91]. Second, validation under real-world conditions remains limited—most reported frameworks are tested on simulated or small-scale testbeds, leading to performance overestimation when deployed in actual microgrids or grid-tied ESS operations [92]. Third, the computational complexity of multi-scale optimization poses significant barriers. Coordinating PV forecasts (spanning seconds to days) with ESS decision-making requires hybrid control frameworks capable of operating across diverse time horizons—something few existing models achieve efficiently [93]. Moreover, current algorithms often fail to integrate battery degradation dynamics and uncertainty quantification, which are crucial for ensuring both reliability and longevity of ESS assets [94]. The interdisciplinary nature of the problem requiring expertise in control theory, optimization, power electronics, and data science has slowed the translation of theoretical advances into standardized operational frameworks. As recent reviews emphasize, bridging this gap will require large-scale open datasets, hybrid physics-informed ML models, and hierarchical control architectures that couple forecasting with real-time decision-making [95].

4.9. Synthesis of Findings

4.9.1. Global Research Trends

Our integrated analysis highlights a rapidly growing research landscape in PV–ESS forecasting, especially after 2018. Bibliometric patterns show that the field's expansion has been accompanied by a shift in focus toward advanced AI and grid-integration themes: recent co-occurrence networks reveal emerging keywords such as "deep learning," "smart grids," "microgrids," and "reinforcement learning," reflecting the rise in data-driven methods and holistic energy management strategies in the post-2018 literature. This surge in publications has, however, outpaced methodological rigor.

4.9.2. Methodological Quality

A qualitative appraisal of the 50 reviewed studies found that only about 60% meet high quality standards, while roughly 40% exhibit moderate-to-low quality, with many lacking external validation and omitting any uncertainty analysis. In practice, this means a significant portion of recent papers report results on the original datasets without independent test sets or confidence intervals, underscoring that the rapid growth of the field has not been matched by consistent rigor in study design.

4.9.3. Aggregated Model Performance

Nonetheless, the evidence confirms that modern machine learning (ML) and deep learning (DL) models deliver substantially higher forecasting accuracy than traditional time-series or physical methods. The systematic review consistently found ML/DL approaches outperform benchmarks like ARIMA or persistence models in highly variable solar scenarios. Correspondingly, our meta-analysis of the small subset of studies reporting comparable metrics yielded a mean R^2 of approximately 0.95 in favor of ML-based forecasts, quantitatively illustrating the high predictive power of these techniques (albeit this average comes from only five studies and thus must be interpreted with caution). Notably, complex hybrid architectures such as CNN–LSTM networks and other deep hybrids integrating convolutional, recurrent, or attention mechanisms tend to achieve the best performance, capturing spatiotemporal patterns more effectively than single-model approaches. These gains in accuracy translate into tangible operational benefits: for example, some studies have reported that coupling ML forecasts with battery control can boost PV energy utilization and efficiency.

4.9.4. Gaps in ML–ESS Integration

Despite the progress, critical gaps and open questions remain. Few studies truly close the loop between forecasting and storage control–indeed, the intersection of PV forecasting with ESS operational management is still largely underexplored. The majority of works evaluate predictive accuracy in isolation, without demonstrating how improved forecasts actually impact real-world ESS performance or economic outcomes. As a result, it remains unclear how much added value these ML-based forecasts provide in practice–most papers stop short of reporting the quantitative benefit of integrating an ESS based on the forecast. Additionally, the field lacks a practical taxonomy or standardization for forecast horizons: studies define forecasting intervals (from very short-term to day-ahead) inconsistently, making it difficult to compare results or establish generalizable best practices. Almost no long-term field validations or comprehensive economic impact studies have been published, so the durability and financial merits of ML-enhanced PV–ESS forecasts remain unproven beyond simulation or short trial periods. In summary, our findings paint a dual-faced picture, on one hand, substantial advances in ML techniques are driving marked improvements in PV forecasting accuracy and hinting at better PV–ESS operations; on the other hand, methodological weaknesses and unresolved integration challenges currently limit the translation of these forecasting gains into fully realized grid benefits. This overarching perspective sets the stage for a deeper discussion on how to interpret the results in context and what steps are needed to address the identified gaps moving forward.

4.10. Limitations

4.10.1. Methodological Limitations

Despite such advances, recurring methodological deficiencies were identified among the analyzed studies. One of the most notable issues is insufficient model validation: many studies do not employ cross-validation or rigorous external testing, leading to

overfitting risks and raising doubts about the generalization of results beyond the original dataset. Indeed, around 40% of the reviewed articles demonstrated moderate to low methodological quality, with the absence of external validation and uncertainty analysis being the most frequent shortcomings. The latter point is critical, as few studies report confidence intervals or probabilistic forecasts—a necessary practice for safely managing ESS under solar variability. Another limitation observed is the heterogeneity in data and evaluation metrics: each study often uses different datasets, forecasting horizons, and evaluation criteria, making objective performance comparisons between methods difficult. Such methodological disparities, combined with the high computational demands of some deep models, hinder the extraction of general conclusions about which approach is most effective under specific conditions or the real quantitative benefit of integrating an ESS based on these forecasts. These findings emphasize the need to strengthen experimental rigor and transparency in this emerging field by establishing standards that enable more reliable study comparisons.

4.10.2. Limitations of the Meta-Analysis

The small sample size of the meta-analysis (only 5 studies) limits the generalizability of our findings. With so few data points, the pooled R^2 estimate is imprecise and must be viewed cautiously. Notably, statistical tests for heterogeneity have very low power when the number of studies is small [96]. Thus, the absence of observed between-study variability (I^2 = 0%) in our results likely reflects insufficient power to detect differences rather than true uniformity across all contexts. In practical terms, we cannot conclude that all studies are genuinely homogeneous in outcome; rather, the meta-analysis may simply be underpowered to reveal existing heterogeneity. Lack of variance data and the use of external weighting introduce methodological weaknesses. Most of the included studies did not report the variance or confidence interval for their R^2 values, precluding the standard inverse-variance weighting approach. As a workaround, we applied externally approximated weights in the random-effects model. This compromise diminishes the statistical rigor of the analysis, because the weights are not optimally reflecting each study's precision. Consequently, the 95% confidence interval for the pooled R^2 had to be calculated under normality assumptions and was extremely wide (even extending beyond the logical 0–1 range of R^2). Such a wide interval underscores the imprecision of the combined estimate and further stresses that the pooled result should be interpreted with caution rather than taken as a precise value.

No sensitivity or robustness analysis was conducted due to the minimal number of studies. In a typical meta-analysis, one would perform a leave-one-out analysis or other sensitivity checks to ensure that no single study unduly influences the overall result. However, with k = 5, removing even one study (20% of the data) would make the analysis extremely unstable, so we did not attempt a leave-one-out procedure. Likewise, no subgroup analyses or meta-regressions were feasible given the small sample. The absence of these robustness checks means we have not formally tested how sensitive the pooled R^2 is to any individual study or assumption. This is an important limitation, and future meta-analyses with a larger study pool should include sensitivity analyses (e.g., leave-one-out or subgroup analyses) to verify the stability of their findings. there are inherent issues with the R^2 metric and its interpretability in our meta-analysis context. R^2 is a bounded measure (restricted to [0, 1]) and is not normally distributed, especially when values are clustered near the upper bound. This characteristic complicates the calculation of standard confidence intervals and the application of parametric significance tests. Indeed, treating R^2 as a continuous normal outcome in our analysis led to a misleading confidence interval (approximately –0.44 to 2.33) that breached the theoretical limits of the metric. Moreover, the R^2 values reported in

the individual studies were all very high, ranging from 0.87 to 1.00. When performance is so close to the maximum possible value, small absolute differences in R^2 may not reflect substantive differences in model performance. For instance, improving from $R^2 = 0.90$ to $R^2 = 0.95$ might not yield a practically significant improvement in forecast accuracy, since both values indicate a very high level of explained variance. Thus, the near-ceiling effect of R^2 in this domain means that differences between models can appear minor on this scale even if they would correspond to non-trivial changes in other error metrics or operational outcomes.

5. Conclusions

This work integrates scientometric analysis, a systematic review (PRISMA 2020), and a focused meta-analysis to elucidate how AI/ML techniques enhance photovoltaic (PV) forecasting when integrated with energy storage systems (ESS). It updates the state of the art through 2025 and provides a specific synthesis on the forecast–operation interaction of ESS, a dimension previously absent in general reviews of ML applied to solar energy. The main contribution lies in (i) thematically updating the field with emerging models (Transformers, probabilistic forecasting, predictive control), (ii) critically comparing approaches (deep networks vs. classical methods; short- vs. long-horizon forecasting) from the standpoint of ESS applicability, and (iii) analyzing synergies and challenges in the joint implementation of forecasting and storage. Moreover, the PRISMA protocol reduced 227 initial records to 50 studies for qualitative synthesis and 5 with comparable R^2 values for meta-analysis, ensuring traceability and relevance of the evaluated corpus.

The evidence converges in showing that ML/DL consistently outperform traditional approaches (ARIMA, persistence) for PV power prediction under high variability. In particular, LSTM, CNN, and hybrid or attention-based (Transformer) architectures achieve the greatest error reductions by capturing complex spatiotemporal dependencies relevant to optimal battery management. The random-effects meta-analysis yielded a combined $R^2 \approx 0.95$ with no heterogeneity ($I^2 = 0\%$), supporting the high average explanatory power of ML/DL across different contexts—albeit with wide confidence intervals due to the lack of reported variances. Operationally, integrating forecasts into ESS control schemes increases self-consumption and efficiency: up to ~87% PV utilization has been documented with forecast-based charging strategies, and on a larger scale, the addition of ESS—supported by reliable forecasting—can increase a project's net value up to fivefold compared to PV without storage.

Nevertheless, methodological gaps persist that nuance the interpretation and transferability of the results: (i) insufficient external validation and scarce reporting of uncertainty or probabilistic outputs; (ii) heterogeneity in data, forecasting horizons, and evaluation metrics that hinder fair comparisons; and (iii) lack of variance or standard deviation data for rigorous quantitative synthesis. The quality assessment indicates that 60% of the studies exhibit low risk of bias, 30% moderate, and 10% serious limitations, with frequent issues being the absence of external validation and uncertainty analysis. The use of JAMOVI/REML provided traceability, but the lack of variance data limited the precision of the combined effect.

Based on these findings, we propose concrete priorities for future research and implementation:

(1) Standardize benchmarks (reference datasets, common horizons, and unified metrics) to fairly assess different models and quantify their benefits for ESS integration.
(2) Incorporate robustness and resilience criteria—e.g., performance under atypical events, missing data, and domain shift—alongside conventional average error metrics.

(3) Close the forecast–action loop through deep reinforcement learning for adaptive charge/discharge management.
(4) Deploy XAI techniques to enhance interpretability and operational acceptance among system operators.
(5) Expand data availability in poorly instrumented sites through controlled synthetic data generation while maintaining bias auditing.

These directions, already outlined in the discussion, will enable a transition from local accuracy improvements to systemic gains in operational safety, battery lifespan, and overall system economics.

We conclude that the transferability of these practices extends to emerging technologies and multi-energy systems: PV–BESS coordination with green hydrogen production and DC/EVCS microgrids benefits from accurate forecasts coupled with intelligent optimization and control, enabling stable operation and improvements in efficiency and cost for H_2/O_2 coproduction and bidirectional charging. Thus, artificial intelligence is consolidated as a key enabler for more precise, safe, and economically efficient PV–ESS integration, while also supporting the development of next-generation flexible and decarbonized energy architectures.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANN	Artificial Neural Network (a type of machine learning model)
ARIMA	Autoregressive Integrated Moving Average (a statistical time series model)
ARIMAX	Autoregressive Integrated Moving Average with Exogenous variables
BESS	Battery Energy Storage System (refers specifically to battery-based ESS)
BMS	Battery Management System
BR	Bayesian Regularization (an algorithm/training method for neural networks)
CNN	Convolutional Neural Network (a type of deep learning model)
DL	Deep Learning
DoD	Depth of Discharge
DRL	Deep Reinforcement Learning
EV	Electric Vehicle
EVCS	Electric Vehicle Charging Station
FFNN	Feed-Forward Neural Network
FL	Fuzzy Logic
FLC	Fuzzy Logic Controller
PV-ESS	Photovoltaic + Energy Storage System
REML	Restricted Maximum Likelihood

UDDS	Urban Dynamometer Driving Schedule
UPQC	Unified Power Quality Conditioner
WT	Wavelet Transform

References

1. Benitez, I.B.; Singh, J.G. A comprehensive review of machine learning applications in forecasting solar PV and wind turbine power output. *J. Electr. Syst. Inf. Technol.* **2025**, *12*, 54. [CrossRef]
2. de Oliveira, J.F.L.; de Mattos Neto, P.S.G.; Siqueira, H.V.; Santos, D.S.d.O., Jr.; Lima, A.R.; Madeiro, F.; Dantas, D.A.P.; Cavalcanti, M.d.M.; Pereira, A.C.; Marinho, M.H.N. Forecasting methods for photovoltaic energy in the scenario of battery energy storage systems: A comprehensive review. *Energies* **2023**, *16*, 6638. [CrossRef]
3. Lalbiakhlua, K.; Deb, S.; Singh, K.R.; Datta, S.; Shuaibu, H.A.; Ustun, T.S. Deep learning based solar forecasting for optimal PV BESS sizing in ultra fast charging stations. *Sci. Rep.* **2025**, *15*, 32392. [CrossRef] [PubMed]
4. Nadeem, A.; Hanif, M.F.; Naveed, M.S.; Hassan, M.T.; Gul, M.; Husnain, N.; Mi, J. AI-Driven precision in solar forecasting: Breakthroughs in machine learning and deep learning. *AIMS Geosci.* **2024**, *10*, 684–734. [CrossRef]
5. Zhao, M.; Xue, B.; Li, B.; Zhu, J.; Song, W. Ensemble learning with support vector machines algorithm for surface roughness prediction in longitudinal vibratory ultrasound-assisted grinding. *Precis. Eng.* **2024**, *88*, 382–400. [CrossRef]
6. Liu, S.; Lv, J.; Lv, P.; Cheng, Y. Deep learning-based electricity theft detection for active distribution networks with high photovoltaic penetration. *Comput. Electr. Eng.* **2025**, *126*, 110551. [CrossRef]
7. Li, H.; Wu, W.; Chen, W.; Zhang, M. RTI-Net: Physics-informed deep learning for photovoltaic power forecasting. *Renew. Energy* **2025**, *256*, 124152. [CrossRef]
8. Xu, S.; Zhao, M.; Hao, Y.; Ji, C. Real-time mooring tension prediction for a floating offshore photovoltaic system using a CNN-Bi-LSTM hybrid neural network. *Ocean. Eng.* **2025**, *340*, 122322. [CrossRef]
9. Taherdoost, H. Deep Learning and Neural Networks: Decision-Making Implications. *Symmetry* **2023**, *15*, 1723. [CrossRef]
10. Rondón-Cordero, V.H.; Montuori, L.; Alcázar-Ortega, M.; Siano, P. Advancements in hybrid and ensemble ML models for energy consumption forecasting: Results and challenges of their applications. *Renew. Sustain. Energy Rev.* **2025**, *224*, 116095. [CrossRef]
11. Fathima, A.H.; Palanisamy, K. Optimization in microgrids with hybrid energy systems—A review. *Renew. Sustain. Energy Rev.* **2015**, *45*, 431–446. [CrossRef]
12. Thomas, J.K.; Crasta, H.R.; Kausthubha, K.; Gowda, C.; Rao, A. Battery monitoring system using machine learning. *J. Energy Storage* **2021**, *40*, 102741. [CrossRef]
13. Das, U.K.; Karmaker, A.K. Optimizing PV power utilization in standalone battery systems with forecast-based charging management strategy. *Glob. Energy Interconnect.* **2025**, *8*, 407–419. [CrossRef]
14. Bampoulas, A.; Pallonetto, F.; Mangina, E.; Finn, D.P. A Bayesian deep-learning framework for assessing the energy flexibility of residential buildings with multicomponent energy systems. *Appl. Energy* **2023**, *348*, 121576. [CrossRef]
15. Khortsriwong, N.; Boonraksa, P.; Boonraksa, T.; Fangsuwannarak, T.; Boonsrirat, A.; Pinthurat, W.; Marungsri, B. Performance of deep learning techniques for forecasting PV power generation: A case study on a 1.5 MWp floating PV power plant. *Energies* **2023**, *16*, 2119. [CrossRef]
16. Yang, Z.; Li, Y.; Feng, X.; Sun, Y.; Pan, R. Hybrid Deep Learning and Reinforcement Learning Framework for Power Prediction and Optimization in Photovoltaic Storage Systems. In Proceedings of the 7th International Conference on Information Technologies and Electrical Engineering, Guangzhou China, 1–3 November 2024; pp. 208–212.
17. Babay, M.-A.; Adar, M.; Chebak, A.; Mabrouki, M. Forecasting green hydrogen production: An assessment of renewable energy systems using deep learning and statistical methods. *Fuel* **2025**, *381*, 133496. [CrossRef]
18. Aksan, F.; Suresh, V.; Janik, P. Optimal capacity and charging scheduling of battery storage through forecasting of photovoltaic power production and electric vehicle charging demand with deep learning models. *Energies* **2024**, *17*, 2718. [CrossRef]
19. Van Eck, N.; Waltman, L. Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics* **2010**, *84*, 523–538. [CrossRef]
20. Aria, M.; Cuccurullo, C. bibliometrix: An R-tool for comprehensive science mapping analysis. *J. Informetr.* **2017**, *11*, 959–975. [CrossRef]
21. Das, D.C.; Roy, A.K.; Sinha, N. GA based frequency controller for solar thermal–diesel–wind hybrid energy generation/energy storage system. *Int. J. Electr. Power Energy Syst.* **2012**, *43*, 262–279. [CrossRef]
22. Dawoud, S.M.; Lin, X.; Okba, M.I. Hybrid renewable microgrid optimization techniques: A review. *Renew. Sustain. Energy Rev.* **2018**, *82*, 2039–2052. [CrossRef]
23. Lee, S.; Choi, D.-H. Federated reinforcement learning for energy management of multiple smart homes with distributed energy resources. *IEEE Trans. Ind. Inform.* **2020**, *18*, 488–497. [CrossRef]
24. Shin, M.; Choi, D.-H.; Kim, J. Cooperative management for PV/ESS-enabled electric vehicle charging stations: A multiagent deep reinforcement learning approach. *IEEE Trans. Ind. Inform.* **2019**, *16*, 3493–3503. [CrossRef]

25. Husein, M.; Chung, I.-Y. Day-ahead solar irradiance forecasting for microgrids using a long short-term memory recurrent neural network: A deep learning approach. *Energies* **2019**, *12*, 1856. [CrossRef]
26. Kow, K.W.; Wong, Y.W.; Rajkumar, R.K.; Rajkumar, R.K. A review on performance of artificial intelligence and conventional method in mitigating PV grid-tied related power quality events. *Renew. Sustain. Energy Rev.* **2016**, *56*, 334–346. [CrossRef]
27. Hu, D.; Ye, Z.; Gao, Y.; Ye, Z.; Peng, Y.; Yu, N. Multi-agent deep reinforcement learning for voltage control with coordinated active and reactive power optimization. *IEEE Trans. Smart Grid* **2022**, *13*, 4873–4886. [CrossRef]
28. Lee, S.; Choi, D.-H. Energy management of smart home with home appliances, energy storage system and electric vehicle: A hierarchical deep reinforcement learning approach. *Sensors* **2020**, *20*, 2157. [CrossRef]
29. Sarita, K.; Kumar, S.; Vardhan, A.S.S.; Elavarasan, R.M.; Saket, R.K.; Shafiullah, G.M.; Hossain, E. Power enhancement with grid stabilization of renewable energy-based generation system using UPQC-FLC-EVA technique. *IEEE Access* **2020**, *8*, 207443–207464. [CrossRef]
30. Lee, S.; Choi, D.-H. Reinforcement learning-based energy management of smart home with rooftop solar photovoltaic system, energy storage system, and home appliances. *Sensors* **2019**, *19*, 3937. [CrossRef]
31. Ali, M.F.; Sheikh, M.R.I.; Biswas, D.; Al Mamun, A.; Hossen, M.J. Optimizing Renewable Energy-Based Grid-Connected Hybrid Microgrid for Residential Applications in Bangladesh: Predictive Modeling for Renewable Energy, Grid Stability and Demand Response Analysis. *Results Eng.* **2025**, *27*, 106997. [CrossRef]
32. Yang, H.X.; Lu, L.; Burnett, J. Weather data and probability analysis of hybrid photovoltaic–wind power generation systems in Hong Kong. *Renew. Energy* **2003**, *28*, 1813–1824. [CrossRef]
33. Alghamdi, H.; Maduabuchi, C.; Okoli, K.; Alobaid, M.; Alghassab, M.; Alsafran, A.S.; Makki, E.; Alkhedher, M. Latest advancements in solar photovoltaic-thermoelectric conversion technologies: Thermal energy storage using phase change materials, machine learning, and 4E analyses. *Int. J. Energy Res.* **2024**, *2024*, 1050785. [CrossRef]
34. Amir, M.; Khan, S.Z. Assessment of renewable energy: Status, challenges, COVID-19 impacts, opportunities, and sustainable energy solutions in Africa. *Energy Built Environ.* **2022**, *3*, 348–362. [CrossRef]
35. Awad, H.; Nassar, Y.F.; Hafez, A.; Sherbiny, M.K.; Ali, A.F.M. Optimal design and economic feasibility of rooftop photovoltaic energy system for Assuit University, Egypt. *Ain Shams Eng. J.* **2022**, *13*, 101599. [CrossRef]
36. Tepe, İ.F.; Demirtaş, M.; Irmak, E.; Bayindir, R. Eco-friendly energy storage systems based demand side management for rural microgrid by dandelion optimization algorithm. In Proceedings of the 2024 13th International Conference on Renewable Energy Research and Applications (ICRERA), Nagasaki, Japan, 9–13 November 2024; pp. 1657–1667.
37. Castillo-Cagigal, M.; Gutiérrez, A.; Monasterio-Huelin, F.; Caamaño-Martín, E.; Masa, D.; Jiménez-Leube, J. A semi-distributed electric demand-side management system with PV generation for self-consumption enhancement. *Energy Convers. Manag.* **2011**, *52*, 2659–2666. [CrossRef]
38. Varma, R.K. *Smart Solar PV Inverters with Advanced Grid Support Functionalities*; John Wiley & Sons: Hoboken, NJ, USA, 2021.
39. Khayyat, M.M.; Sami, B. Energy community management based on artificial intelligence for the implementation of renewable energy systems in smart homes. *Electronics* **2024**, *13*, 380. [CrossRef]
40. Gopinathan, N.; Gupta, D.; Shanmugam, P.K.; Nersisson, R.; Gabriel, G.A.; Daniel, G.E. Deep-cycle battery sizing and strategic battery management in solar-EV systems using UDDS drive cycles with real-time environment. *Eng. Res. Express* **2025**, *7*, 35350. [CrossRef]
41. Elweddad, M.A.; Guneser, M.T. Intelligent energy management and prediction of micro grid operation based on machine learning algorithms and genetic algorithm. *Int. J. Renew. Energy Res.* **2022**, *12*, 2002–2014. [CrossRef]
42. Agustono, I. State of Charge Prediction of Lead Acid Battery Using Transformer Neural Network for Solar Smart Dome 4.0. 2022. Available online: https://engrxiv.org/preprint/view/2606/version/3774 (accessed on 6 October 2025).
43. Almughram, O.; Abdullah ben Slama, S.; Zafar, B.A. A reinforcement learning approach for integrating an intelligent home energy management system with a vehicle-to-home unit. *Appl. Sci.* **2023**, *13*, 5539. [CrossRef]
44. Seidu, I.; Salisu, S.; Mas'ud, A.A.; Musa, U.; Jabire, A.H.; Muhammad-Sukki, F.; Mas'ud, I.A. An Overview of Current Optimization Approaches for Hybrid Energy Systems Combining Solar Photovoltaic and Wind Technologies. *Energy Sci. Eng.* **2025**, *13*, 4633–4655. [CrossRef]
45. Sarin, C.R.; Mani, G.; Stonier, A.A.; Peter, G.; Kumaresan, P.; Ganji, V. An extensive critique on expert system control in solar photovoltaic dominated microgrids. *IET Renew. Power Gener.* **2025**, *19*, e12875.
46. Odoi-Yorke, F.; Baah, B.; Kissi-Boateng, R.; Opoku, R.; Effah, F.B. Unlocking the potential of redundant energy from solar photovoltaic systems: Navigating global research trends, innovations, and future directions. *Energy Rep.* **2025**, *13*, 6362–6383. [CrossRef]
47. Nwokolo, S.C.; Obiwulu, A.U.; Okonkwo, P.C.; Orji, R.; Alsenani, T.R.; Mansir, I.B.; Orji, C. Machine learning and physics-based model hybridization to assess the impact of climate change on single-and dual-axis tracking solar-concentrated photovoltaic systems. *Phys. Chem. Earth Parts A/B/C* **2025**, *138*, 103881. [CrossRef]

48. Pratap Singh, A.; Kumar, Y.; Sawle, Y.; Alotaibi, M.A.; Malik, H.; Pedro García Márquez, F. Development of artificial Intelligence-Based adaptive vehicle to grid and grid to vehicle controller for electric vehicle charging station. *Ain Shams Eng. J.* **2024**, *15*, 102937. [CrossRef]
49. Manoj, D.; Josh, F.T. A comprehensive review on optimization and artificial intelligence algorithms for effective battery management in EVs. *Int. J. Electr. Electron. Eng. Telecommun.* **2023**, *12*, 334–341. [CrossRef]
50. Mwakitalima, I.J.; Rizwan, M.; Kumar, N. Integrating Solar Photovoltaic Power Source and Biogas Energy-Based System for Increasing Access to Electricity in Rural Areas of Tanzania. *Int. J. Photoenergy* **2023**, *2023*, 7950699. [CrossRef]
51. Di Santo, K.G.; Di Santo, S.G.; Monaro, R.M.; Saidel, M.A. Active demand side management for households in smart grids using optimization and artificial intelligence. *Measurement* **2018**, *115*, 152–161. [CrossRef]
52. Kaewdornhan, N.; Chatthaworn, R. Predictive energy management for microgrid using multi-agent deep deterministic policy gradient with random sampling. *IEEE Access* **2024**, *12*, 95071–95090. [CrossRef]
53. Manlapaz, J.G.C.; Reyes, D.M.; Buensuceso, C.P.L.; Peña, R.A.S.; Parocha, R.C.; Macabebe, E.Q.B. Optimization and simulation of a grid-connected PV system using load forecasting methods: A case study of a university building. *IOP Conf. Ser. Earth Environ. Sci.* **2023**, *1199*, 12006. [CrossRef]
54. Khadeeja, F.P.; Radha, M.; Vanitha, G.; Nirmala, D.M.; Mahendiran, R.; Vishnu, S.S. Daily Solar Power Prediction Using Machine Learning: A Model Wise Comparative Study. 2025. Available online: https://www.horizonepublishing.com/index.php/PST/article/view/8861 (accessed on 6 October 2025).
55. Habib, M.A.; Hossain, M.J. Advanced feature engineering in microgrid PV forecasting: A fast computing and data-driven hybrid modeling framework. *Renew. Energy* **2024**, *235*, 121258. [CrossRef]
56. Khouili, O.; Hanine, M.; Louzazni, M.; Flores, M.A.L.; Villena, E.G.; Ashraf, I. Evaluating the impact of deep learning approaches on solar and photovoltaic power forecasting: A systematic review. *Energy Strateg. Rev.* **2025**, *59*, 101735. [CrossRef]
57. Bera, S.; Chakraborty, S.; Kumar, D.; Ali, N.; Lehtonen, M. Optimal deep learning based aggregation of TCLs in an inverter fed stand-alone microgrid for voltage unbalance mitigation. *Electr. Power Syst. Res.* **2022**, *210*, 108178. [CrossRef]
58. Babu, R.M.; Islam, M.S.; Basher, E. Forecasting and optimization of a residential off-grid solar photovoltaic-battery energy storage system. *Ecol. Eng. Environ. Technol.* **2025**, *26*, 109–120. [CrossRef] [PubMed]
59. Rokonuzzaman, M.; Rahman, S.; Hannan, M.A.; Mishu, M.K.; Tan, W.-S.; Rahman, K.S.; Pasupuleti, J.; Amin, N. Levenberg-Marquardt algorithm-based solar PV energy integrated internet of home energy management system. *Appl. Energy* **2025**, *378*, 124407. [CrossRef]
60. Mishra, M.; Nayak, J.; Naik, B.; Abraham, A. Deep learning in electrical utility industry: A comprehensive review of a decade of research. *Eng. Appl. Artif. Intell.* **2020**, *96*, 104000. [CrossRef]
61. Buonanno, A.; Caputo, G.; Balog, I.; Fabozzi, S.; Adinolfi, G.; Pascarella, F.; Leanza, G.; Graditi, G.; Valenti, M. Machine learning and weather model combination for PV production forecasting. *Energies* **2024**, *17*, 2203. [CrossRef]
62. Torres, N.N.S.; Diaz, V.N.S.; Ando, O.H.; Ledesma, J.J.G. Analysis of the Technical Feasibility of Using Artificial Intelligence for Smoothing Active Power in a Photovoltaic System Connected to the Power System. *Brazilian Arch. Biol. Technol.* **2021**, *64*, e21210196. [CrossRef]
63. Srilakshmi, K.; Sujatha, C.N.; Balachandran, P.K.; Mihet-Popa, L.; Kumar, N.U. Optimal design of an artificial intelligence controller for solar-battery integrated UPQC in three phase distribution networks. *Sustainability* **2022**, *14*, 13992. [CrossRef]
64. Sarathkumar, T.V.; Goswami, A.K.; Khan, B.; Shoush, K.A.; Ghoneim, S.S.M.; Ghaly, R.N.R. Forecasting of virtual power plant generating and energy arbitrage economics in the electricity market using machine learning approach. *Sci. Rep.* **2025**, *15*, 3812. [CrossRef]
65. Kim, J.; Obregon, J.; Park, H.; Jung, J.-Y. Multi-step photovoltaic power forecasting using transformer and recurrent neural networks. *Renew. Sustain. Energy Rev.* **2024**, *200*, 114479. [CrossRef]
66. Prabawa, P.; Choi, D.-H. Safe deep reinforcement learning-assisted two-stage energy management for active power distribution networks with hydrogen fueling stations. *Appl. Energy* **2024**, *375*, 124170. [CrossRef]
67. Touzani, S.; Prakash, A.K.; Wang, Z.; Agarwal, S.; Pritoni, M.; Kiran, M.; Brown, R.; Granderson, J. Controlling distributed energy resources via deep reinforcement learning for load flexibility and energy efficiency. *Appl. Energy* **2021**, *304*, 117733. [CrossRef]
68. Li, D.; Guo, S.; Wang, J.; Li, Y.; Sun, C.; Qiao, G.; Chaomurilige; Ding, Y. Interdependent design and operation of solar photovoltaics and battery energy storage for economically viable decarbonisation of local energy systems. *Energy AI* **2025**, *20*, 100505. [CrossRef]
69. Sherri, M.; Tadepalli, S.; Otto, S.; Lenneman, A.; Babcock, N. Optimizing fault detection in battery energy storage systems through data-driven modeling. *J. Energy Storage* **2025**, *121*, 116466. [CrossRef]
70. Shanmugapriya, V.; Vidyasagar, S.; Raju, D.K. Post-fault voltage recovery and voltage instability assessment of DC microgrid with Deep Transfer-learning Convolution Neural Network. *Electr. Power Syst. Res.* **2025**, *239*, 111234. [CrossRef]
71. Singh, A.K.; Kumar, R.; Chaturvedi, D.K.; Ibraheem; Sharma, G.; Bokoro, P.N.; Kumar, R. Application of Twisting Controller and Modified Pufferfish Optimization Algorithm for Power Management in a Solar PV System with Electric-Vehicle and Load-Demand Integration. *Energies* **2025**, *18*, 3785. [CrossRef]

72. Shern, S.J.; Sarker, M.T.; Ramasamy, G.; Thiagarajah, S.P.; Al Farid, F.; Suganthi, S.T. Artificial Intelligence-Based Electric Vehicle Smart Charging System in Malaysia. *World Electr. Veh. J.* **2024**, *15*, 440. [CrossRef]
73. Wang, N.; Guo, Y.; Liu, L.; Shao, S. Numerical assessment and optimization of photovoltaic-based hydrogen-oxygen Co-production energy system: A machine learning and multi-objective strategy. *Renew. Energy* **2024**, *227*, 120483. [CrossRef]
74. Zhang, L.; Zhu, J.; Zhang, D.; Liu, Y. An incremental photovoltaic power prediction method considering concept drift and privacy protection. *Appl. Energy* **2023**, *351*, 121919. [CrossRef]
75. Miraftabzadeh, S.M.; Colombo, C.G.; Longo, M.; Foiadelli, F. A day-ahead photovoltaic power prediction via transfer learning and deep neural networks. *Forecasting* **2023**, *5*, 213–228. [CrossRef]
76. Besnard, Q.; Ragot, N. Continual Learning for Time Series Forecasting: A First Survey. *Eng. Proc.* **2024**, *68*, 49.
77. Parthasarathy, C.; Laaksonen, H.; Redondo-Iglesias, E.; Pelissier, S. Aging aware adaptive control of Li-ion battery energy storage system for flexibility services provision. *J. Energy Storage* **2023**, *57*, 106268. [CrossRef]
78. Sattarzadeh, S.; Padisala, S.K.; Shi, Y.; Mishra, P.P.; Smith, K.; Dey, S. Feedback-based fault-tolerant and health-adaptive optimal charging of batteries. *Appl. Energy* **2023**, *343*, 121187. [CrossRef]
79. Li, M.; Xu, W.; Zhang, S.; Liu, L.; Hussain, A.; Hu, E.; Zhang, J.; Mao, Z.; Chen, Z. State of health estimation and battery management: A review of health indicators, models and machine learning. *Materials* **2025**, *18*, 145. [CrossRef] [PubMed]
80. Schwartz, R.; Dodge, J.; Smith, N.A.; Etzioni, O. Green ai. *Commun. ACM* **2020**, *63*, 54–63. [CrossRef]
81. Greg, J. Quantization and Pruning Strategies for Energy-Efficient TinyML Models. 2025. Available online: https://www.researchgate.net/publication/395994840_Quantization_and_Pruning_Strategies_for_Energy-Efficient_TinyML_Models (accessed on 6 October 2025).
82. Popovic, G.; Spalevic, Z.; Jovanovic, L.; Zivkovic, M.; Stosic, L.; Bacanin, N. Optimizing lightweight recurrent networks for solar forecasting in tinyml: Modified metaheuristics and legal implications. *Energies* **2024**, *18*, 105. [CrossRef]
83. Lee, M.; Park, J.; Na, S.-I.; Choi, H.S.; Bu, B.-S.; Kim, J. An analysis of battery degradation in the integrated energy storage system with solar photovoltaic generation. *Electronics* **2020**, *9*, 701. [CrossRef]
84. Keerthisinghe, C.; Mickelson, E.; Kirschen, D.S.; Shih, N.; Gibson, S. Improved PV forecasts for capacity firming. *IEEE Access* **2020**, *8*, 152173–152182. [CrossRef]
85. Schade, C.; Egging-Bratseth, R. Battery degradation: Impact on economic dispatch. *Energy Storage* **2024**, *6*, e588. [CrossRef]
86. Grane, M.; Cornejo, M.; Hesse, H.; Jossen, A. Accounting for Subsystem Aging Variability in Battery Energy Storage System Optimization. *arXiv* **2025**, arXiv:2507.04813. [CrossRef]
87. Wang, Y.; Zhou, Z.; Botterud, A.; Zhang, K.; Ding, Q. Stochastic coordinated operation of wind and battery energy storage system considering battery degradation. *J. Mod. Power Syst. Clean. Energy* **2016**, *4*, 581–592. [CrossRef]
88. Wolfs, P.; Emami, K.; Lin, Y.; Palmer, E. Load forecasting for diurnal management of community battery systems. *J. Mod. Power Syst. Clean. Energy* **2018**, *6*, 215–222. [CrossRef]
89. Iliadis, P.; Ntomalis, S.; Atsonios, K.; Nesiadis, A.; Nikolopoulos, N.; Grammelis, P. Energy management and techno-economic assessment of a predictive battery storage system applying a load levelling operational strategy in island systems. *Int. J. Energy Res.* **2021**, *45*, 2709–2727. [CrossRef]
90. Cao, J.; Harrold, D.; Fan, Z.; Morstyn, T.; Healey, D.; Li, K. Deep reinforcement learning-based energy storage arbitrage with accurate lithium-ion battery degradation model. *IEEE Trans. Smart Grid* **2020**, *11*, 4513–4521. [CrossRef]
91. Alcañiz, A.; Grzebyk, D.; Ziar, H.; Isabella, O. Trends and gaps in photovoltaic power forecasting with machine learning. *Energy Rep.* **2023**, *9*, 447–471. [CrossRef]
92. Cantillo-Luna, S.; Moreno-Chuquen, R.; Celeita, D.; Anders, G. Deep and machine learning models to forecast photovoltaic power generation. *Energies* **2023**, *16*, 4097. [CrossRef]
93. Sohani, A.; Sayyaadi, H.; Cornaro, C.; Shahverdian, M.H.; Pierro, M.; Moser, D.; Karimi, N.; Doranehgard, M.H.; Li, L.K.B. Using machine learning in photovoltaics to create smarter and cleaner energy generation systems: A comprehensive review. *J. Clean. Prod.* **2022**, *364*, 132701. [CrossRef]
94. Mohamad Radzi, P.N.L.; Akhter, M.N.; Mekhilef, S.; Mohamed Shah, N. Review on the application of photovoltaic forecasting using machine learning for very short-to long-term forecasting. *Sustainability* **2023**, *15*, 2942. [CrossRef]
95. Di Leo, P.; Ciocia, A.; Malgaroli, G.; Spertino, F. Advancements and Challenges in Photovoltaic Power Forecasting: A Comprehensive Review. *Energies* **2025**, *18*, 2108. [CrossRef]
96. von Hippel, P.T. The heterogeneity statistic I 2 can be biased in small meta-analyses. *BMC Med. Res. Methodol.* **2015**, *15*, 35. [CrossRef]

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